



US012384508B2

(12) **United States Patent**
Ernst et al.

(10) **Patent No.:** **US 12,384,508 B2**
(45) **Date of Patent:** ***Aug. 12, 2025**

(54) **EXHAUST HANDLING SYSTEMS FOR MARINE VESSELS AND RELATED METHODS**

(71) Applicant: **MARATHON PETROLEUM COMPANY LP**, Findlay, OH (US)

(72) Inventors: **Stephen Daniel Ernst**, Findlay, OH (US); **James A. Tomlinson**, Findlay, OH (US)

(73) Assignee: **MARATHON PETROLEUM COMPANY LP**, Findlay, OH (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

This patent is subject to a terminal disclaimer.

(21) Appl. No.: **18/660,606**

(22) Filed: **May 10, 2024**

(65) **Prior Publication Data**

US 2024/0286726 A1 Aug. 29, 2024

Related U.S. Application Data

(63) Continuation of application No. 18/369,485, filed on Sep. 18, 2023, now Pat. No. 12,043,361.
(Continued)

(51) **Int. Cl.**
B63H 21/32 (2006.01)
B63H 20/24 (2006.01)
(Continued)

(52) **U.S. Cl.**
CPC **B63H 21/32** (2013.01); **F01N 13/004** (2013.01); **F01N 13/085** (2013.01); **B63H 20/24** (2013.01);
(Continued)

(58) **Field of Classification Search**
CPC combination set(s) only.
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

2,626,627 A 1/1953 Jung et al.
2,864,252 A 12/1958 Schaschl
(Continued)

FOREIGN PATENT DOCUMENTS

AU 2010241217 11/2010
AU 2013202839 5/2014
(Continued)

OTHER PUBLICATIONS

Annex to Form PCT/ISA/206, Communication Relating to the Results of the Partial International Search for international application No. PCT/US2024/021101 mailed Aug. 13, 2024.

(Continued)

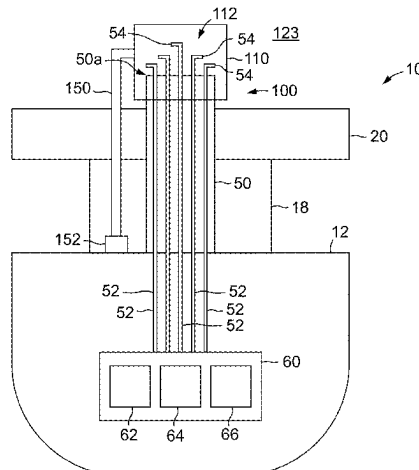
Primary Examiner — Binh Q Tran

(74) *Attorney, Agent, or Firm* — Ray Quinney & Nebeker P.C.; Paul N. Taylor

(57) **ABSTRACT**

An embodiment of an exhaust handling system for a marine vessel includes a cap connected to a top end portion of an exhaust stack of the marine vessel to form an enclosure at least partially surrounding an outlet of an exhaust pipe extending through the exhaust stack. In addition, the exhaust handling system includes a collection pipe in fluid communication with the cap such that the collection pipe is to receive exhaust from the enclosure, and a coupling connected to the collection pipe that is to connect to an exhaust cleaning assembly. The exhaust cleaning system includes a tank to receive the exhaust. The cap at least partially defines a first flow path for the exhaust that extends from the enclosure to the atmosphere. The collection pipe at least partially defines a second flow path for the exhaust that

(Continued)



extends from the enclosure to the coupling via the collection pipe.

24 Claims, 14 Drawing Sheets

Related U.S. Application Data

- (60) Provisional application No. 63/488,574, filed on Mar. 6, 2023, provisional application No. 63/485,886, filed on Feb. 18, 2023.

(51) Int. Cl.

B63H 21/34 (2006.01)
F01N 1/08 (2006.01)
F01N 1/16 (2006.01)
F01N 3/023 (2006.01)
F01N 3/28 (2006.01)
F01N 13/00 (2010.01)
F01N 13/08 (2010.01)
F01N 13/18 (2010.01)

(52) U.S. Cl.

CPC **B63H 20/245** (2013.01); **B63H 21/34** (2013.01); **F01N 1/085** (2013.01); **F01N 1/161** (2013.01); **F01N 1/163** (2013.01); **F01N 3/0235** (2013.01); **F01N 3/0236** (2013.01); **F01N 3/2892** (2013.01); **F01N 13/082** (2013.01); **F01N 13/1816** (2013.01); **F01N 13/1822** (2013.01); **F01N 2240/22** (2013.01); **F01N 2490/12** (2013.01); **F01N 2490/16** (2013.01); **F01N 2490/18** (2013.01); **F01N 2490/20** (2013.01); **F01N 2590/02** (2013.01); **F01N 2590/022** (2013.01)

(56) References Cited

U.S. PATENT DOCUMENTS

3,087,311 A 4/1963 Rousseau
 3,303,525 A 2/1967 Peoples
 3,398,071 A 8/1968 Bagno
 3,504,686 A 4/1970 Cooper et al.
 3,593,555 A 7/1971 Grosko
 3,608,869 A 9/1971 Woodle
 3,672,180 A 6/1972 Davis
 3,725,669 A 4/1973 Tatum
 3,807,433 A 4/1974 Byrd
 3,809,113 A 5/1974 Grove
 3,814,148 A 6/1974 Wostl
 3,925,592 A 12/1975 Webb
 3,961,493 A 6/1976 Nolan, Jr.
 4,010,779 A 3/1977 Pollock et al.
 4,073,303 A 2/1978 Foley, Jr.
 4,109,677 A 8/1978 Burnside
 4,202,351 A 5/1980 Biche
 4,229,064 A 10/1980 Vetter et al.
 4,242,533 A 12/1980 Cott
 4,289,163 A 9/1981 Pierson
 4,294,378 A 10/1981 Rabinovich
 4,315,602 A 2/1982 Kubacak
 4,320,775 A 3/1982 Stirling et al.
 4,357,576 A 11/1982 Hickam et al.
 4,420,008 A 12/1983 Shu
 4,457,037 A 7/1984 Rylander
 4,481,474 A 11/1984 Gerrit
 4,488,570 A 12/1984 Jiskoot
 4,630,685 A 12/1986 Huck et al.
 4,690,587 A 9/1987 Petter
 4,744,305 A 5/1988 Lim et al.
 4,784,324 A 11/1988 DeWitt
 4,788,093 A 11/1988 Murata et al.

4,794,331 A 12/1988 Schweitzer, Jr.
 4,848,082 A 7/1989 Takahashi
 4,897,226 A 1/1990 Hoyle et al.
 4,904,932 A 2/1990 Schweitzer, Jr.
 4,964,732 A 10/1990 Cadeo et al.
 5,050,064 A 9/1991 Mayhew
 5,072,380 A 12/1991 Randelman et al.
 5,095,977 A 3/1992 Ford
 5,129,432 A 7/1992 Dugger
 5,191,537 A 3/1993 Edge
 5,305,631 A 4/1994 Whited
 5,367,882 A 11/1994 Lievens et al.
 5,383,243 A 1/1995 Thacker
 5,423,607 A 6/1995 Jones
 5,469,830 A 11/1995 Gonzalez
 5,516,967 A 5/1996 Pandey
 5,533,912 A 7/1996 Fillingner
 5,562,133 A 10/1996 Mitchell
 5,595,709 A 1/1997 Klemp
 5,603,360 A 2/1997 Teel
 5,627,749 A 5/1997 Waterman et al.
 5,628,351 A 5/1997 Ramsey, Jr. et al.
 5,660,602 A 8/1997 Collier, Jr.
 5,661,623 A 8/1997 McDonald
 5,783,916 A 7/1998 Blackburn
 5,814,982 A 9/1998 Thompson et al.
 5,832,967 A 11/1998 Andersson
 5,873,916 A 2/1999 Cemenska et al.
 5,887,974 A 3/1999 Pozniak
 5,895,347 A 4/1999 Doyle
 5,906,648 A 5/1999 Zoratti et al.
 5,906,877 A 5/1999 Popper et al.
 5,939,166 A 8/1999 Cheng et al.
 5,962,774 A 10/1999 Mowry
 5,973,593 A 10/1999 Botella
 5,993,054 A 11/1999 Tan et al.
 6,022,421 A 2/2000 Bath
 6,050,844 A 4/2000 Johnson
 6,065,903 A 5/2000 Doyle
 6,077,340 A 6/2000 Doyle
 6,077,418 A 6/2000 Iseri et al.
 6,098,601 A 8/2000 Reddy
 6,111,021 A 8/2000 Nakahama et al.
 6,149,351 A 11/2000 Doyle
 6,186,193 B1 2/2001 Phallen et al.
 6,220,747 B1 4/2001 Gosselin
 6,243,483 B1 6/2001 Petrou et al.
 6,328,877 B1 12/2001 Bushman
 6,333,374 B1 12/2001 Chen
 6,346,813 B1 2/2002 Kleinberg
 6,383,237 B1 5/2002 Langer et al.
 6,427,384 B1 8/2002 Davis, Jr.
 6,478,353 B1 11/2002 Barrozo
 6,679,302 B1 1/2004 Mattingly et al.
 6,719,921 B2 4/2004 Steinberger et al.
 6,799,883 B1 10/2004 Urquhart et al.
 6,834,531 B2 12/2004 Rust
 6,840,292 B2 1/2005 Hart et al.
 6,851,916 B2 2/2005 Schmidt
 6,980,647 B1 12/2005 Daugherty et al.
 6,987,877 B2 1/2006 Paz-Pujalt et al.
 7,032,629 B1 4/2006 Mattingly et al.
 7,091,421 B2 8/2006 Kukita et al.
 7,121,040 B2 10/2006 Wiese
 7,168,464 B2 1/2007 Diggins
 7,186,321 B2 3/2007 Benham
 7,258,710 B2 8/2007 Caro et al.
 7,275,366 B2 10/2007 Powell et al.
 7,294,913 B2 11/2007 Fischer et al.
 7,385,681 B2 6/2008 Ninomiya et al.
 7,444,996 B2 11/2008 Potier
 7,459,067 B2 12/2008 Dunn et al.
 7,564,540 B2 7/2009 Paulson
 7,631,671 B2 12/2009 Mattingly et al.
 7,685,943 B2 3/2010 Caro et al.
 7,729,561 B1 6/2010 Boland et al.
 7,749,308 B2 7/2010 McCully
 7,810,988 B2 10/2010 Kamimura et al.
 7,815,744 B2 10/2010 Abney et al.

US 12,384,508 B2

Page 3

(56)

References Cited

U.S. PATENT DOCUMENTS

7,832,338 B2	11/2010	Caro et al.	10,962,437 B1	3/2021	Nottrott et al.
7,879,204 B2	2/2011	Funahashi	10,970,927 B2	4/2021	Sharp
8,075,651 B2 *	12/2011	Caro B08B 15/02 440/89 R	10,990,114 B1	4/2021	Miller
8,282,265 B2	10/2012	Breithaupt	10,997,707 B1	5/2021	Katz et al.
8,299,811 B2	10/2012	Wing	11,010,608 B2	5/2021	Adam et al.
8,312,584 B2	11/2012	Hodde	11,027,304 B2	6/2021	Donaldson
8,327,631 B2	12/2012	Caro et al.	11,066,971 B1	7/2021	Sharp
8,368,405 B2	2/2013	Siebens	11,112,308 B2	9/2021	Kreitinger et al.
8,376,432 B1	2/2013	Halger et al.	11,125,391 B2	9/2021	Al Khowaiter et al.
8,402,746 B2	3/2013	Powell et al.	11,132,008 B2	9/2021	Miller
8,413,484 B2	4/2013	Lubkowitz	11,164,406 B2	11/2021	Meroux et al.
8,414,781 B2	4/2013	Berard	11,215,095 B2	1/2022	Tonsich
8,577,518 B2	11/2013	Linden et al.	11,221,107 B2	1/2022	Du et al.
8,597,380 B2	12/2013	Buchanan	11,247,184 B2	2/2022	Miller
8,616,760 B2	12/2013	Williams et al.	11,320,519 B2	5/2022	Koivuranta
8,632,359 B2	1/2014	Grimm	11,325,687 B1	5/2022	Sharp
8,647,162 B2	2/2014	Henriksson et al.	11,332,070 B2	5/2022	Holden et al.
8,748,677 B2	6/2014	Buchanan	11,345,455 B2 *	5/2022	Sharp B63B 1/04
8,808,415 B2	8/2014	Caro et al.	11,371,405 B2	6/2022	Sharp
8,912,924 B2	12/2014	Scofield et al.	11,416,012 B2	8/2022	Miller
8,979,982 B2	3/2015	Jordan et al.	11,428,600 B2	8/2022	Dankers et al.
9,038,855 B2	5/2015	Lurcott et al.	11,428,622 B2	8/2022	Borin et al.
9,089,806 B2	7/2015	Powell et al.	11,441,088 B2	9/2022	Robbins
9,162,944 B2	10/2015	Bennett et al.	11,447,877 B1	9/2022	Ell
9,175,235 B2	11/2015	Kastner	11,559,774 B2	1/2023	Miller
9,222,480 B2	12/2015	Younes et al.	11,565,221 B2	1/2023	Miller
9,310,016 B2	4/2016	Hodde	11,578,631 B2	2/2023	Tonsich
9,329,066 B2	5/2016	Skarping	11,578,638 B2	2/2023	Thobe
9,363,462 B2	6/2016	Yoel	11,578,836 B2	2/2023	Thobe
9,388,350 B2	7/2016	Buchanan	11,596,910 B2	3/2023	Miller
9,497,956 B2	11/2016	Dubose	11,607,654 B2	3/2023	Miller
9,518,693 B2	12/2016	Hodde	11,655,748 B1	5/2023	Thobe
9,530,121 B2	12/2016	Brauer et al.	11,655,940 B2	5/2023	Thobe
9,550,247 B2	1/2017	Smith	11,662,750 B2	5/2023	Miller
9,643,135 B1	5/2017	Mazzei et al.	11,686,070 B1	6/2023	Jordan et al.
9,945,333 B2	4/2018	Kopinsky	11,687,858 B2	6/2023	Rentz
10,001,240 B1	6/2018	Dray et al.	11,715,950 B2	8/2023	Miller et al.
10,012,340 B1	7/2018	Dray et al.	11,720,526 B2	8/2023	Miller et al.
10,024,768 B1	7/2018	Johnsen	11,739,679 B2	8/2023	Thobe
10,094,508 B1	10/2018	Dray et al.	11,752,472 B2	9/2023	Miller
10,132,220 B2	11/2018	Tonsich	11,754,225 B2	9/2023	Thobe
10,134,042 B1	11/2018	Prasad et al.	11,761,366 B2	9/2023	Thobe
10,168,255 B1	1/2019	Johnsen	11,774,042 B2	10/2023	Thobe
10,196,243 B1	2/2019	Wells	11,789,453 B2	10/2023	Chowdhary
10,197,206 B1	2/2019	Dray et al.	11,794,153 B2	10/2023	Miller
10,223,596 B1	3/2019	Edwards et al.	11,807,945 B2	11/2023	Ell
10,247,643 B1	4/2019	Johnsen	11,808,013 B1	11/2023	Jordan et al.
10,261,279 B1	4/2019	Potter	11,815,227 B2	11/2023	Thobe
10,287,940 B2	5/2019	Tonsich	11,842,538 B2	12/2023	Saxena
10,345,221 B1	7/2019	Silverman	11,920,504 B2	3/2024	Thobe
10,364,718 B2	7/2019	Eddaoudi et al.	11,965,317 B2	4/2024	Jordan
10,386,260 B2	8/2019	Dudek	11,987,336 B2	5/2024	Sharp
10,405,475 B2	9/2019	Goda	11,988,336 B2	5/2024	Thobe
10,408,377 B1	9/2019	Dray et al.	12,000,538 B2	6/2024	Thobe
10,422,260 B2	9/2019	Tonsich	12,006,014 B1 *	6/2024	Ernst F01N 3/00
10,486,946 B1	11/2019	Wells	12,011,697 B2	6/2024	Miller
10,501,385 B1	12/2019	Buckner et al.	12,012,082 B1	6/2024	Pittman, Jr.
10,513,957 B2	12/2019	Sharp	12,012,883 B2	6/2024	Thobe
10,550,748 B2	2/2020	Sharp	12,043,361 B1 *	7/2024	Ernst B63H 21/32
10,563,555 B2	2/2020	Hamad	12,043,905 B2	7/2024	Ell
10,570,581 B2	2/2020	Faivre	12,043,906 B2	7/2024	Ell
10,605,144 B2 *	3/2020	Kobayashi F01N 3/18	12,066,843 B2	8/2024	Miller
10,619,539 B2	4/2020	Tonsich	12,071,205 B2	8/2024	Sharp
10,633,830 B2	4/2020	Shibamori	12,087,002 B1	9/2024	Miller et al.
10,655,774 B1	5/2020	Dray et al.	12,109,543 B2	10/2024	Miller
10,657,443 B2	5/2020	Araujo et al.	12,128,369 B2	10/2024	Miller
10,688,686 B2	6/2020	Fadhel et al.	12,129,559 B2	10/2024	Ell
10,756,459 B2	8/2020	Jongsma	12,163,625 B2	12/2024	Thobe
10,833,434 B1	11/2020	Tassell, Jr.	12,180,597 B2	12/2024	Ell
10,900,399 B2	1/2021	Tonsich	2001/0013517 A1	8/2001	Hart
10,943,357 B2	3/2021	Badawy et al.	2002/0014068 A1	2/2002	Mittricker et al.
10,948,471 B1	3/2021	MacMullin et al.	2002/0178806 A1	12/2002	Valentine
10,953,960 B1	3/2021	Sharp	2002/0185180 A1	12/2002	Smith
10,960,958 B2	3/2021	Sharp	2003/0041518 A1	3/2003	Wallace et al.
			2003/0121481 A1	7/2003	Dodd et al.
			2003/0158630 A1	8/2003	Pham et al.
			2003/0167660 A1	9/2003	Kondou
			2003/0178994 A1	9/2003	Hurlimann et al.
			2003/0188536 A1	10/2003	Mittricker

(56)

References Cited

U.S. PATENT DOCUMENTS

2003/0197622	A1	10/2003	Reynard et al.	2015/0081165	A1	3/2015	Yamashita et al.
2003/0227821	A1	12/2003	Bae et al.	2015/0144468	A1	5/2015	Skolozdra
2004/0057334	A1	3/2004	Wilmer et al.	2015/0183102	A1	7/2015	Breschi et al.
2004/0058597	A1	3/2004	Matsuda	2015/0198518	A1	7/2015	Borin et al.
2004/0067126	A1	4/2004	Schmidt	2015/0244087	A1	8/2015	Wing
2004/0125688	A1	7/2004	Kelley et al.	2015/0269288	A1	9/2015	Moore
2004/0249105	A1	12/2004	Nolte et al.	2015/0323119	A1	11/2015	Giunta
2004/0251313	A1	12/2004	Burgess	2016/0071059	A1	3/2016	Petering
2004/0265653	A1	12/2004	Buechi et al.	2016/0091467	A1	3/2016	Morris
2005/0007450	A1	1/2005	Hill et al.	2016/0139355	A1	5/2016	Petersen
2005/0058016	A1	3/2005	Smith et al.	2016/0169098	A1	6/2016	Makita
2005/0146437	A1	7/2005	Ward	2016/0169436	A1	6/2016	Sander et al.
2005/0150820	A1	7/2005	Guo	2016/0175634	A1	6/2016	Radian
2005/0154132	A1	7/2005	Hakuta et al.	2016/0238194	A1	8/2016	Adler et al.
2005/0176482	A1	8/2005	Raisinghani et al.	2016/0252650	A1	9/2016	Hirst, Sr.
2005/0284333	A1	12/2005	Falkiewicz	2016/0363249	A1	12/2016	Disher
2006/0125826	A1	6/2006	Lubkowitz	2016/0369930	A1	12/2016	Poe et al.
2006/0263283	A1	11/2006	Egan	2017/0051472	A1	2/2017	Mochimaru
2006/0278304	A1	12/2006	Mattingly et al.	2017/0088401	A1	3/2017	Clements et al.
2007/0175511	A1	8/2007	Doerr	2017/0122174	A1	5/2017	Patel
2007/0181083	A1	8/2007	Fulton	2017/0131728	A1	5/2017	Lambert et al.
2008/0092625	A1	4/2008	Hinnrichs	2017/0140237	A1	5/2017	Voeller et al.
2008/0113884	A1	5/2008	Campbell et al.	2017/0158303	A1	6/2017	Michaelis et al.
2008/0115834	A1	5/2008	Geoffrion et al.	2017/0180012	A1	6/2017	Tingler et al.
2008/0149481	A1	6/2008	Hurt	2017/0248569	A1	8/2017	Lambert et al.
2008/0283083	A1	11/2008	Piao	2017/0253737	A1	9/2017	Auld et al.
2009/0009308	A1	1/2009	Date et al.	2017/0253738	A1	9/2017	Auld et al.
2009/0107111	A1	4/2009	Oliver	2017/0253806	A1	9/2017	Auld et al.
2009/0154288	A1	6/2009	Heathman	2017/0254481	A1	9/2017	Cadogan et al.
2009/0175738	A1	7/2009	Shaimi	2017/0259229	A1	9/2017	Chou et al.
2009/0183498	A1	7/2009	Uchida et al.	2017/0306428	A1	10/2017	Helgason et al.
2009/0188565	A1	7/2009	Satake	2017/0326474	A1	11/2017	Olovsson
2009/0197489	A1	8/2009	Caro	2017/0367346	A1	12/2017	Rees et al.
2009/0278839	A1	11/2009	Geis	2018/0002617	A1	1/2018	Umansky et al.
2010/0031825	A1	2/2010	Kemp	2018/0003116	A1	1/2018	Fersman et al.
2010/0049410	A1	2/2010	McKee	2018/0037452	A1	2/2018	Gray et al.
2010/0058666	A1	3/2010	Kim	2018/0080356	A1*	3/2018	Fukui F01N 3/05
2010/0175316	A1	7/2010	Kubacak	2018/0098137	A1	4/2018	Saha
2010/0198775	A1	8/2010	Rousselle	2018/0119882	A1	5/2018	Allidieres et al.
2011/0265449	A1*	11/2011	Powell B08B 15/00 60/272	2018/0143734	A1	5/2018	Ochenas et al.
2012/0027298	A1	2/2012	Dow	2018/0186528	A1	7/2018	Tonn
2012/0092835	A1	4/2012	Miller	2018/0218214	A1	8/2018	Pestun
2012/0143560	A1	6/2012	Tabet et al.	2018/0223202	A1	8/2018	Fransham et al.
2012/0153042	A1	6/2012	Oedekoven	2018/0245313	A1	8/2018	Shibamori et al.
2012/0185220	A1	7/2012	Shippen	2018/0259064	A1	9/2018	McLemore
2012/0276379	A1	11/2012	Daniels et al.	2018/0312391	A1	11/2018	Borg
2012/0304625	A1	12/2012	Daikoku	2019/0016963	A1	1/2019	Auld et al.
2013/0035824	A1	2/2013	Nakamura	2019/0121373	A1	4/2019	Panigrahi
2013/0048094	A1	2/2013	Ballantyne	2019/0367732	A1	5/2019	Helgason et al.
2013/0062258	A1	3/2013	Ophus	2019/0264592	A1	8/2019	Tonsich
2013/0125323	A1	5/2013	Henderson	2019/0270500	A1	9/2019	Hamaoka
2013/0176656	A1	7/2013	Kaisser	2019/0295189	A1	9/2019	Strasser
2013/0186671	A1	7/2013	Theis	2019/0338203	A1	11/2019	Umansky et al.
2013/0201025	A1	8/2013	Kamalakkannan et al.	2019/0359899	A1	11/2019	Umansky et al.
2013/0245524	A1	9/2013	Schofield	2019/0362147	A1	11/2019	Adam
2013/0293884	A1	11/2013	Lee et al.	2019/0136060	A1	12/2019	Helgason et al.
2013/0299500	A1	11/2013	McKinnon	2019/0368054	A1	12/2019	Gummow et al.
2013/0317959	A1	11/2013	Joos	2019/0368156	A1	12/2019	Faivre
2014/0002639	A1	1/2014	Cheben et al.	2020/0033252	A1	1/2020	Borin et al.
2014/0008926	A1	1/2014	Allen	2020/0118413	A1	4/2020	Kanukurthy et al.
2014/0062490	A1	3/2014	Neuman et al.	2020/0232191	A1	7/2020	Prior
2014/0090379	A1	4/2014	Powell et al.	2020/0235559	A1	7/2020	Neuenschwander
2014/0121622	A1	5/2014	Jackson et al.	2020/0240302	A1	7/2020	Tonsich
2014/0133824	A1	5/2014	Yoel	2020/0240588	A1	7/2020	Al Khawaiter
2014/0158616	A1	6/2014	Govind et al.	2020/0245551	A1	8/2020	Hoffman et al.
2014/0158632	A1	6/2014	Govind et al.	2020/0245552	A1	8/2020	Hoffman et al.
2014/0171538	A1	6/2014	Daniels et al.	2020/0245553	A1	8/2020	Hoffman et al.
2014/0172513	A1	6/2014	MacLean	2020/0292445	A1	9/2020	Morimoto
2014/0176344	A1	6/2014	Littlestar	2020/0325742	A1	10/2020	Astudillo et al.
2014/0190691	A1	7/2014	Vinegar	2021/0053011	A1	2/2021	Sugiyama et al.
2014/0194657	A1	7/2014	Wadhwa et al.	2021/0062697	A1	3/2021	Yokoyama et al.
2014/0299039	A1	10/2014	Trollux	2021/0073692	A1	3/2021	Saha et al.
2014/0345370	A1	11/2014	Marotta	2021/0076006	A1	3/2021	O'Neill et al.
2014/0356707	A1	12/2014	Kwon et al.	2021/0095380	A1	4/2021	Borin et al.
				2021/0123211	A1	4/2021	Miller et al.
				2021/0138399	A1	5/2021	Yokoyama et al.
				2021/0192938	A1	6/2021	Doerr et al.
				2021/0197151	A1	7/2021	Miller
				2021/0207772	A1	7/2021	Norton et al.

(56)	References Cited			CA	2642295	1/2010
	U.S. PATENT DOCUMENTS			CA	2736733	10/2011
				CA	2958443	4/2017
				CA	2995532	4/2017
2021/0215925	A1	7/2021	Kim et al.	CA	2916141	6/2017
2021/0216852	A1	7/2021	Reece et al.	CN	2092562	1/1992
2021/0232163	A1	7/2021	Miller	CN	200958686	10/2007
2021/0232741	A1	7/2021	Ogiso et al.	CN	100348970	11/2007
2021/0348281	A1	11/2021	da Costa	CN	102997052	3/2013
2021/0362637	A1	11/2021	Hanis et al.	CN	202898548	U 4/2013
2021/0381920	A1	12/2021	Jacobsz et al.	CN	103106764	5/2013
2022/0001969	A1 *	1/2022	Pugnetti	CN	103497804	1/2014
2022/0010707	A1	1/2022	Sharma et al.	CN	102997061	5/2015
2022/0048606	A1	2/2022	Singh	CN	204824775	12/2015
2022/0081261	A1	3/2022	Karbassi	CN	205640252	10/2016
2022/0087099	A1	3/2022	Hoffman et al.	CN	104372350	B 2/2017
2022/0154427	A1	5/2022	Misaki	CN	106764463	1/2019
2022/0178114	A1	6/2022	Takahama	CN	208306600	1/2019
2022/0183208	A1	6/2022	Sibley	CN	110513604	11/2019
2022/0186470	A1	6/2022	Chiba et al.	CN	210176958	3/2020
2022/0213603	A1	7/2022	Al Eid et al.	CN	111537157	8/2020
2022/0221368	A1	7/2022	Bergeron	CN	114001278	2/2022
2022/0228345	A1	7/2022	Case et al.	CN	113719746	11/2022
2022/0266196	A1	8/2022	Sharp	CN	115325460	A 11/2022
2022/0282651	A1	9/2022	Reynolds et al.	CN	114877263	4/2023
2022/0290411	A1	9/2022	Anahara et al.	CN	220458389	U 2/2024
2022/0343229	A1	10/2022	Gruber et al.	EP	2458573	5/2012
2022/0401899	A1	12/2022	Miller	EP	2586304	A1 5/2013
2022/0404272	A1	12/2022	Kendall et al.	EP	2602609	6/2013
2022/0405870	A1	12/2022	Conway	EP	3076461	10/2016
2023/0012038	A1	1/2023	Iyer	EP	3101411	12/2016
2023/0012673	A1	1/2023	Fukuyama et al.	EP	3112011	1/2017
2023/0015077	A1	1/2023	Kim	EP	2994626	1/2018
2023/0061824	A1	3/2023	Elli	EP	3285759	2/2018
2023/0078852	A1	3/2023	Campbell et al.	ES	2398302	3/2013
2023/0129513	A1	4/2023	Miller	FR	2388762	11/1978
2023/0259080	A1	8/2023	Whitehart et al.	FR	2689241	10/1993
2023/0259088	A1	8/2023	Borup et al.	GB	1179978	2/1970
2023/0332532	A1	10/2023	Thobe	GB	2097687	11/1982
2023/0333577	A1	10/2023	Miller	GB	2545207	6/2017
2023/0333578	A1	10/2023	Miller	GB	2559149	4/2022
2023/0338897	A1 *	10/2023	Bednar	IN	202141001384	1/2021
2023/0341092	A1	10/2023	Thobe	IT	201900008235	12/2020
2023/0347303	A1	11/2023	Miller	JP	2004125039	4/2004
2023/0358023	A1	11/2023	Jordan et al.	JP	2007204023	8/2007
2023/0360151	A1	11/2023	Swamy	JP	2008097832	4/2008
2023/0366510	A1	11/2023	Thobe	JP	2012002159	11/2014
2023/0383416	A1	11/2023	Elli	JP	2016078893	5/2016
2023/0383417	A1	11/2023	Elli	KR	20110010316	2/2011
2023/0383418	A1	11/2023	Elli	KR	20130038986	4/2013
2023/0392536	A1	12/2023	Thobe	KR	20200007444	A 1/2020
2023/0398473	A1	12/2023	Tonsich	KR	102129951	7/2020
2023/0399817	A1	12/2023	Jordan	KR	102169280	10/2020
2023/0399818	A1	12/2023	Jordan	KR	102281640	7/2021
2023/0407488	A1	12/2023	Elli	RU	2760879	12/2021
2023/0415106	A1	12/2023	Miller	RU	217534	U1 4/2023
2024/0003016	A1	1/2024	Elli	WO	1996006685	5/1996
2024/0060189	A1	2/2024	Elli	WO	1997006004	2/1997
2024/0071073	A1	2/2024	Rajora	WO	1997006298	2/1997
2024/0141506	A1	5/2024	Elli	WO	1998003711	1/1998
2024/0166492	A1	5/2024	Thobe	WO	199917606	4/1999
2024/0209988	A1	6/2024	Thobe	WO	2000063108	10/2000
2024/0217498	A1	7/2024	Pittman, Jr.	WO	2002030551	4/2002
2024/0255102	A1	8/2024	Thobe	WO	2002082181	10/2002
2024/0269626	A1	8/2024	Miller	WO	2003003002	1/2003
2024/0271556	A1	8/2024	Thobe	WO	2003066423	8/2003
2024/0278762	A1	8/2024	Pittman, Jr.	WO	2004003293	1/2004
2024/0278894	A1	8/2024	Ernst	WO	2004092307	10/2004
2024/0301811	A1	9/2024	Thobe	WO	2005018300	3/2005
2024/0327992	A1	10/2024	Elli	WO	2007107652	9/2007
2024/0327993	A1	10/2024	Elli	WO	2007112335	10/2007
2024/0367762	A1	11/2024	Sharp	WO	2007149851	12/2007
	FOREIGN PATENT DOCUMENTS			WO	2009013544	1/2009
				WO	2009055024	4/2009
				WO	2010002274	1/2010
AU	2009202785	B	11/2014	WO	2010042704	4/2010
CA	2447358		4/2005	WO	2010103260	9/2010
CA	2702151		10/2007	WO	2011127535	10/2011
CA	2637421		1/2010	WO	2013112274	8/2013

(56)

References Cited

FOREIGN PATENT DOCUMENTS

WO	2014089443	6/2014
WO	2014173672	10/2014
WO	2015061868	5/2015
WO	2015153607	10/2015
WO	2016004107	1/2016
WO	2016026043	2/2016
WO	2016146404	9/2016
WO	2017074985	5/2017
WO	2017083778	5/2017
WO	2017087731	5/2017
WO	2017152269	9/2017
WO	2018005141	1/2018
WO	2018/031520	2/2018
WO	2018102378	6/2018
WO	2019/226348	5/2019
WO	2020044026	3/2020
WO	2020118020	6/2020
WO	2020132632	6/2020
WO	2020223803	11/2020
WO	2020237112	11/2020
WO	2021062563	4/2021
WO	2021100054	5/2021
WO	2022043197	3/2022
WO	2022126092	6/2022
WO	2022149501	7/2022
WO	2022269052	12/2022
WO	2023287276	1/2023
WO	2023021769	2/2023
WO	2023038579	3/2023
WO	2023060350	4/2023
WO	2023137304	7/2023
WO	2023164683	8/2023
ZA	9606765	2/1998
ZA	200610366	1/2008

OTHER PUBLICATIONS

Sadovnychiy, Sergiy et al. "Geographical information system applications for pipeline right of way aerial surveillance", International Conference on Geographical Information Systems Theory, Applications and Management, vol. 2, SCITEPRESS, 2017.

Paschal, Kayla, "Utility Right of Way Management: Potential for Expanded Integrated Vegetation Management in California", 2014. International Search Report and Written Opinion for international application No. PCT/US2024/021101 mailed on Oct. 9, 2024.

Alexandrakis et al., "Marine Transportation for Carbon Capture and Sequestration (CCS)", Department of Civil and Environmental Engineering, Thesis, Massachusetts Institute of Technology, Jun. 2010.

Datta et al., "Advancing carbon management through the global commoditization of CO₂: the case for dual-use LNG-CO₂ shipping", Carbon Management, 2020, vol. 11, No. 6, 611-630.

Ibitoye et al., "Poster Abstract: A Convolutional Neural Network Based Solution for Pipeline Leak Detection", School of Information Technology, Carleton University, Ottawa, Canada, Nov. 2019.

Intelliview, "Thermal Imaging Provides Early Leak Detection in Oil and Gas Pipelines", Petro Industry News, www.Petro-Online.com, Aug./Sep. 2018.

Southwest Research Institute, "Methane Leak Detection", 2021.

Masterduct, "Case Studies: High temp marine grade ship engine exhaust fume hose", retrieved at <https://www.masterduct.com/CaseStudies/HighTempShipEngineExhaustFumeHose.aspx>.

ACTI, "Advanced Maritime Emissions Control System (AMECS)", retrieved at <https://sustainableworldports.org/wp-content/uploads/presentation-on-AMECS.pdf>.

Lloyd's Register, Using technology to trace the carbon intensity of sustainable marine fuels, Feb. 15, 2023.

BORIN Manufacturing, Inc., DART for Test Station, Above Ground Remote Monitoring, Feb. 11, 2021.

BORIN Manufacturing, Inc., COMMANCHE Remote Monitoring and Control System, Mar. 24, 2017.

BORIN Manufacturing, Inc., DART for Rectifiers, Remote Monitoring and Control System, Nov. 1, 2017.

BORIN Manufacturing, Inc., STEALTH 2 Solid-State Reference Electrode for Buried and Concrete Service, Aug. 7, 2015.

BORIN Manufacturing, Inc., STEALTH 3, Nov. 10, 2016.

BORIN Manufacturing, Inc., STEALTH Reference Electrodes, Feb. 4, 2016.

BORIN Manufacturing, Inc., STEALTH Solid-State Reference Electrodes, Nov. 8, 2016.

BORIN Manufacturing, Inc., STEALTH Reference Electrodes, Oct. 10, 2017.

BORIN Manufacturing, Inc., 'Miracle half-cell', Palladium: BORIN's new reference electrode chemistry, Aug. 13, 2014.

BORIN Manufacturing, Inc., Street Dart, For Test Station, Ground Level Remote Monitoring, Mar. 2017.

Skelton et al., Onboard Refueling Vapor Recovery Systems Analysis of Widespread Use, Nescaum, Boston MA, Aug. 20, 2007.

Membrane Technology and Research, Inc., Gasoline Vapor Recovery, 2018.

Jordan Technologies, Aereon, Recovering More Vapor = Increased Profits, 2015.

EPFL, Capturing CO₂ from trucks and reducing their emissions by 90%, Dec. 23, 2019.

Sharma, Shivom et al., Carbon Dioxide Capture from Internal Combustion Engine Exhaust Using Temperature Swing Adsorption, Front. Energy Res., Sec. Carbon Capture, Utilization and Storage, Dec. 16, 2019.

Information Disclosure Declaration by Kyle E. Miller, Dec. 18, 2020.

Cott Manufacturing Company, FinkLet®/FinkPlate® Cathodic Protection Test Stations, Wayback Machine, May 22, 2000.

Neutrik XXR-2 Xx Series, <https://www.parts-express.com/Neutrik-XXR-2-XX-Series-Color-Coding-Ring-Red>, 2022.

Hou, Qingmin, An FBG Strain Sensor-Based NPW Method for Natural Gas Pipeline Leakage Detection, Hindawi, Mathematical Problems in Engineering, vol. 2021, Article ID 5548503, pp. 1-8.

International Search Report and Written Opinion for international application No. PCT/US2024/021099 mailed on Aug. 2, 2024.

* cited by examiner

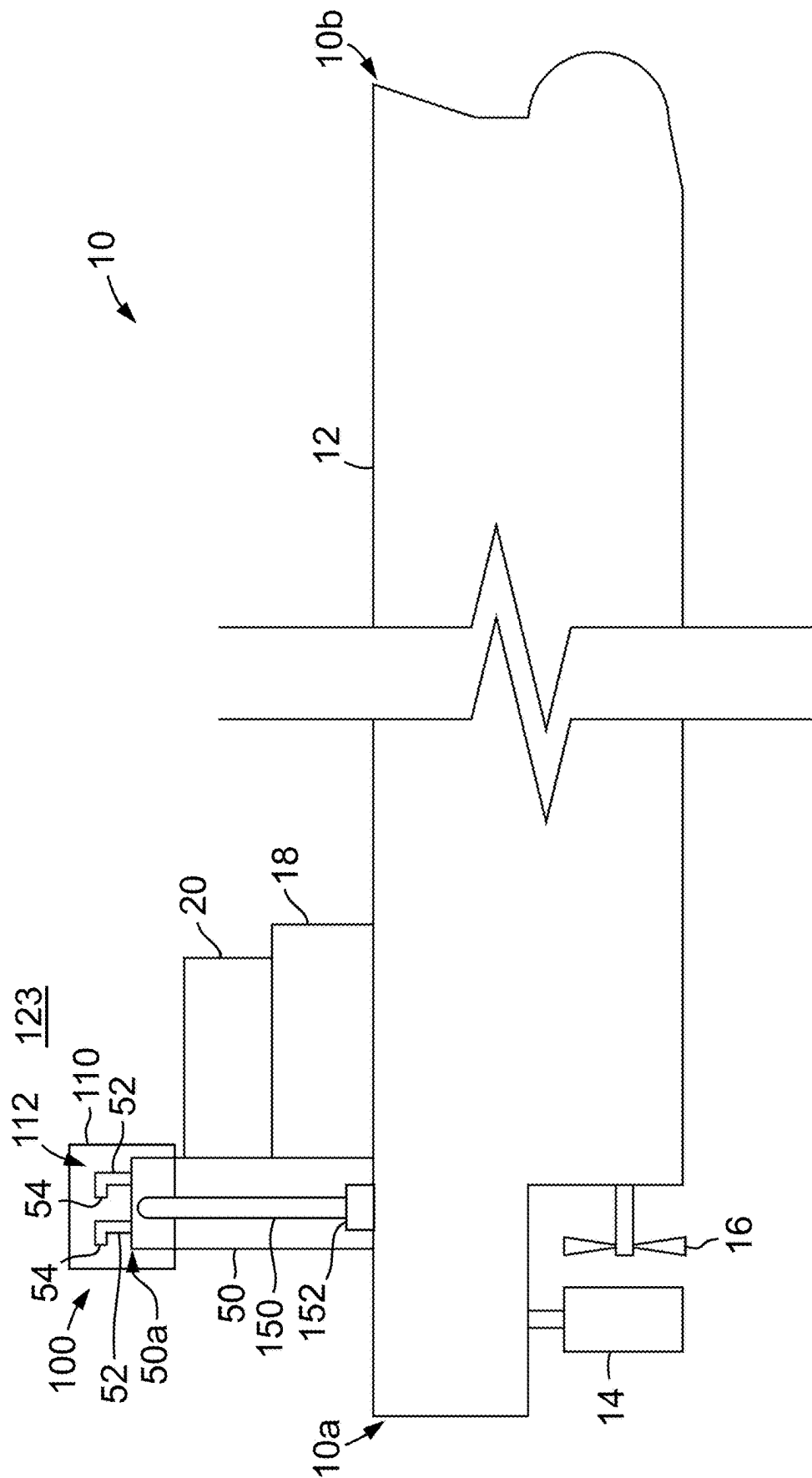


Fig. 1

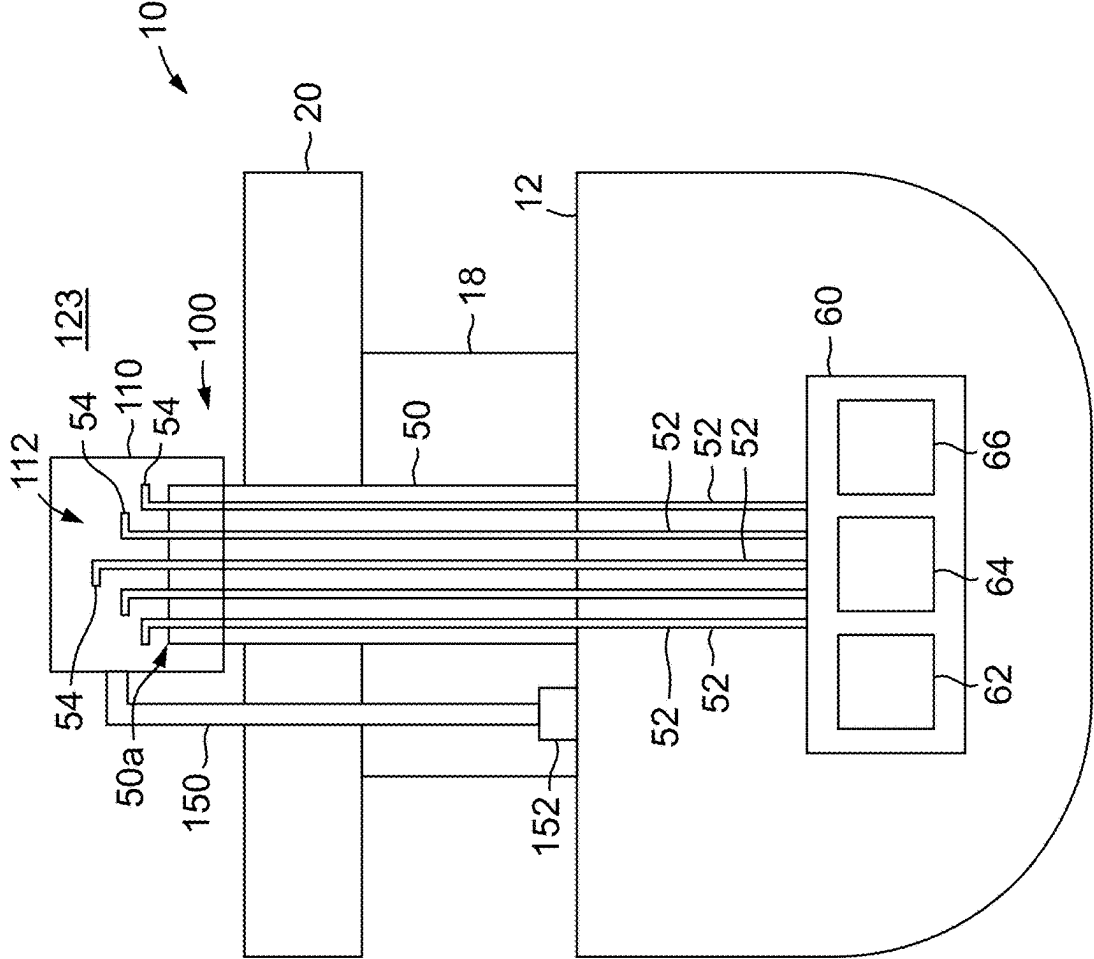


FIG. 2

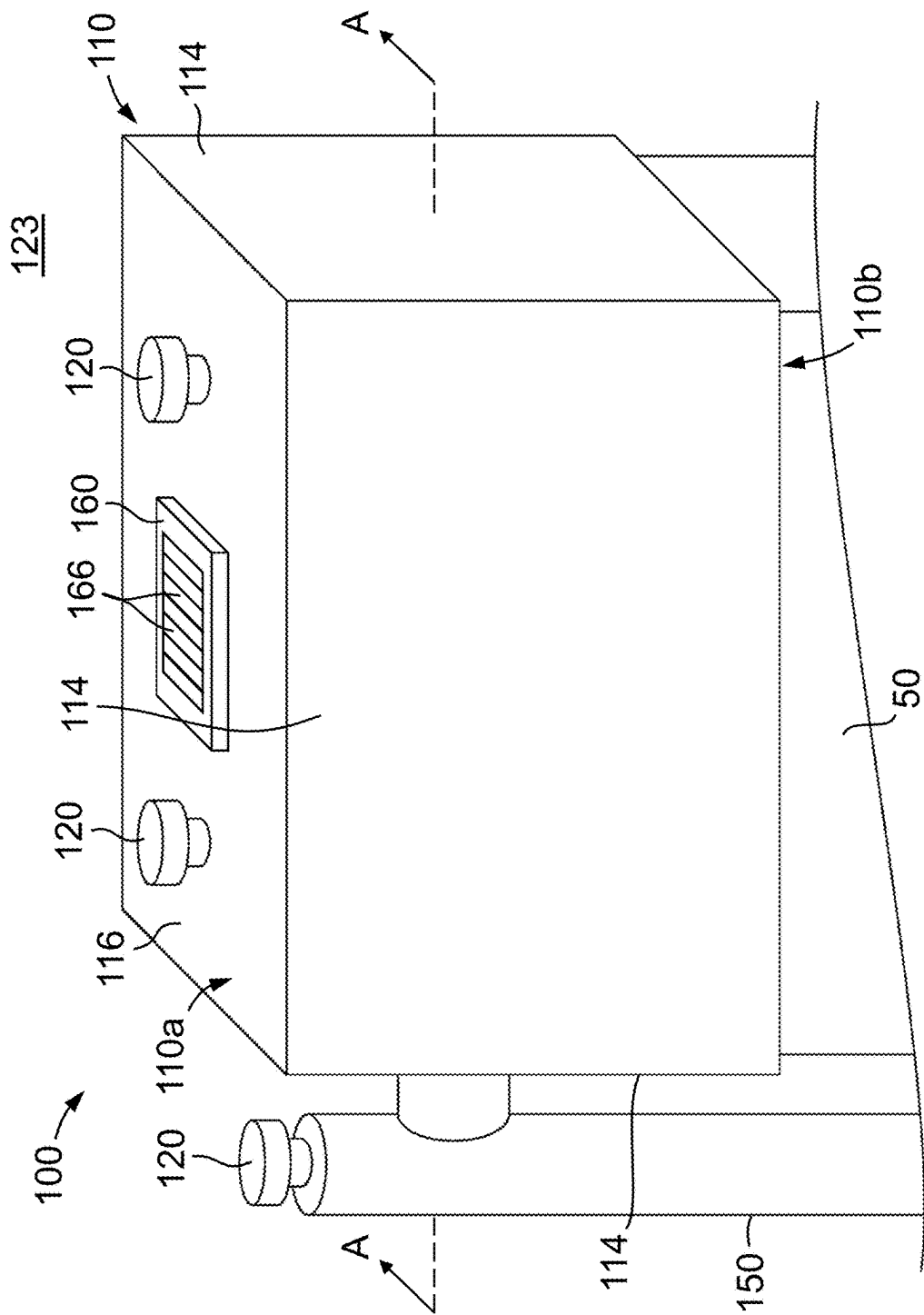


FIG. 3

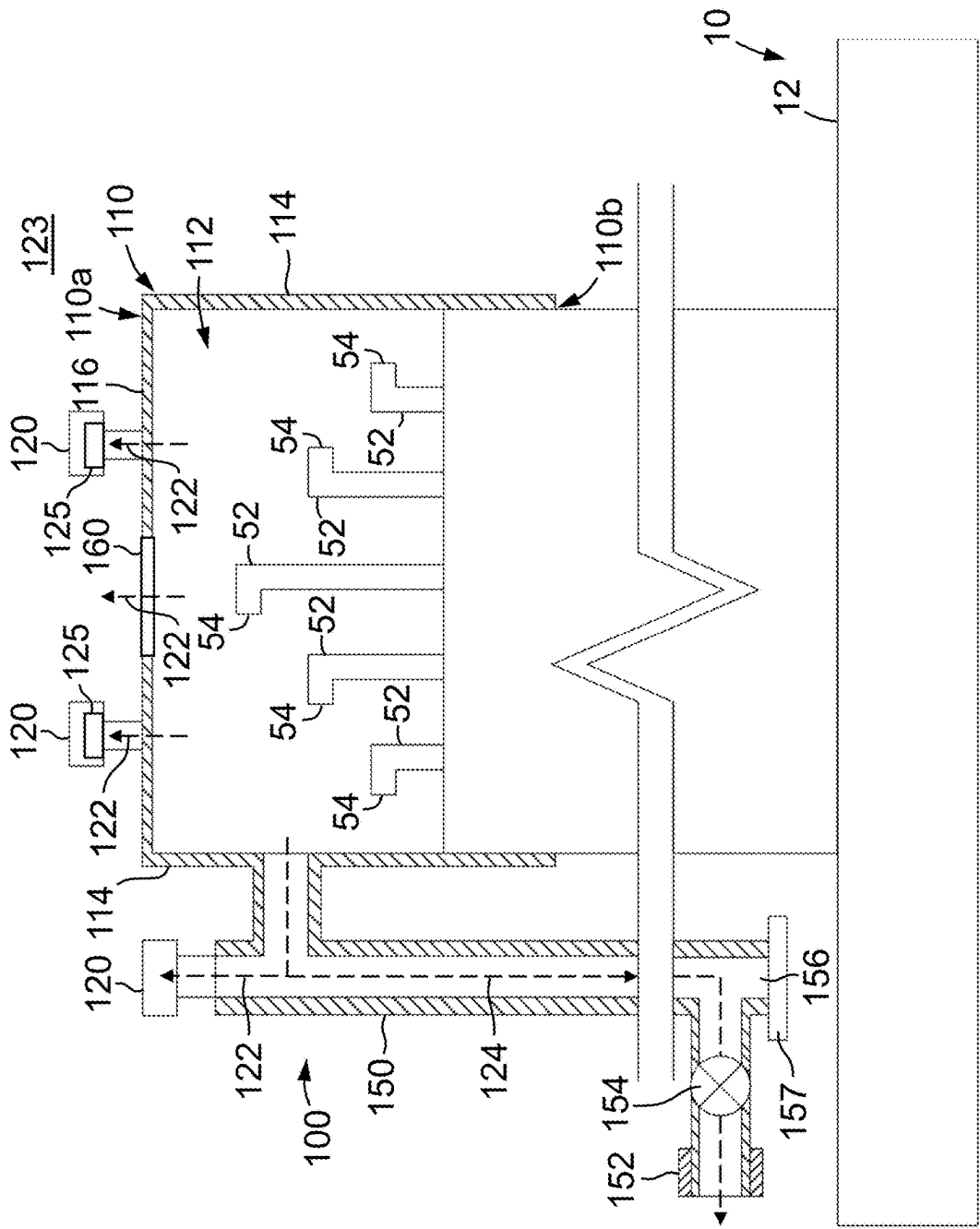


FIG. 4

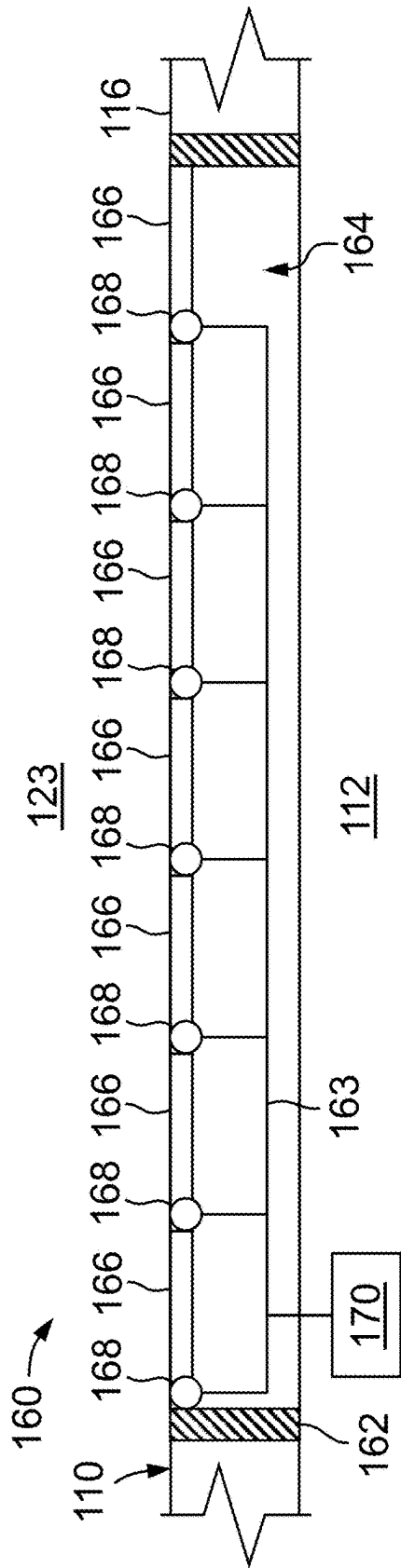


FIG. 5

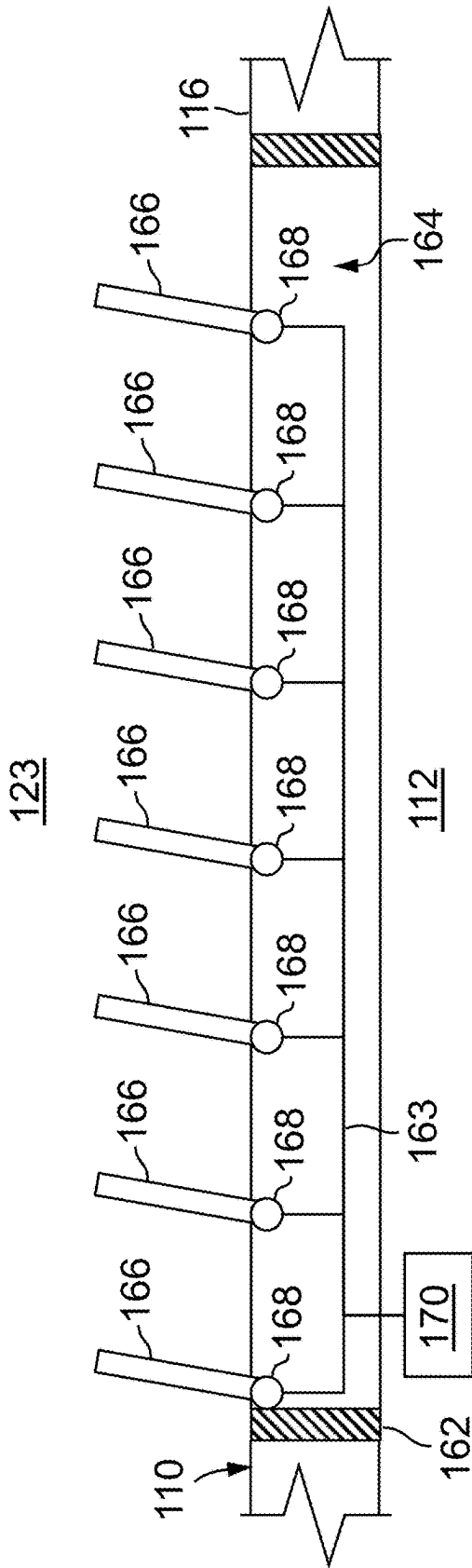


FIG. 6

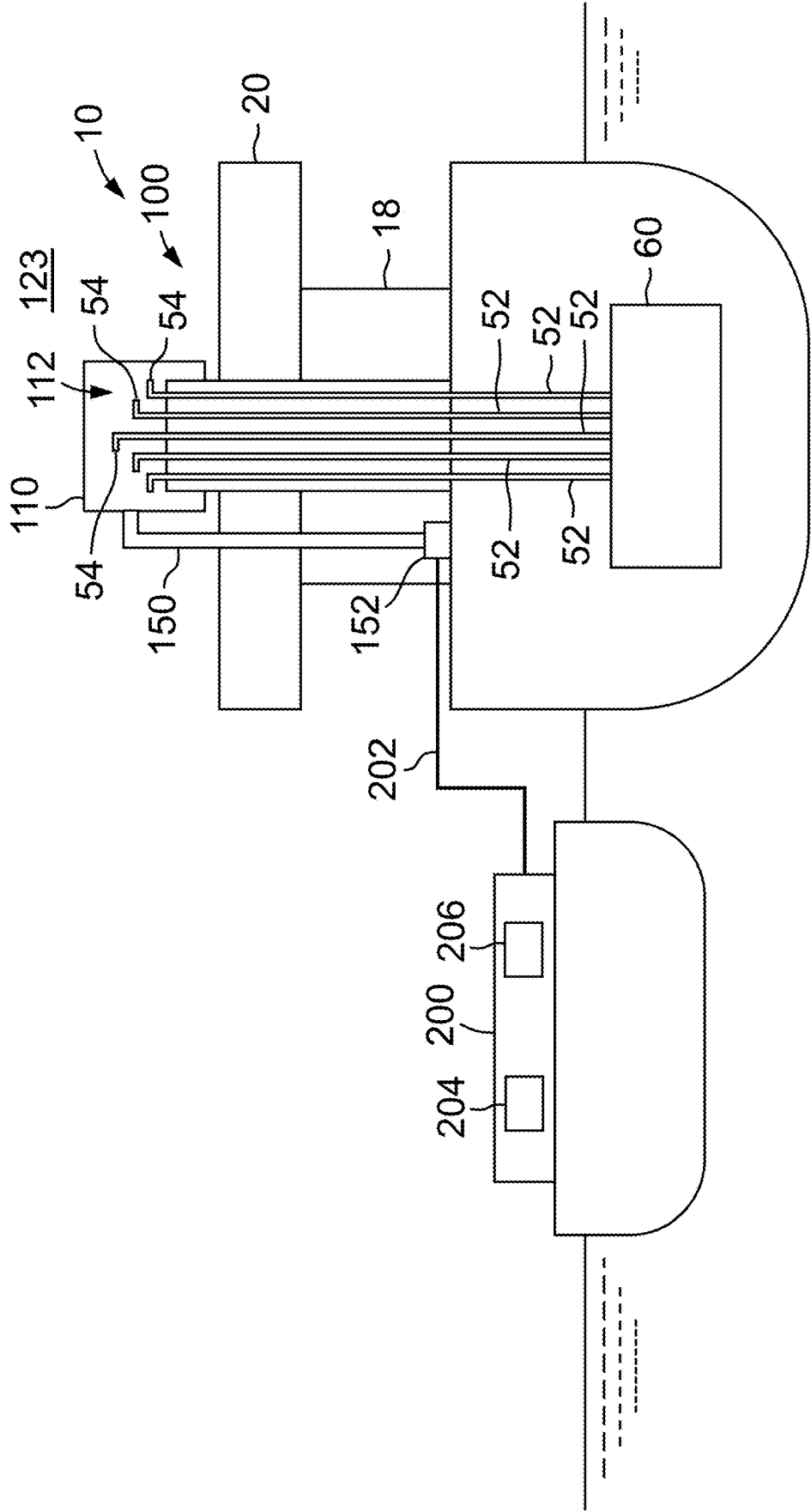


FIG. 7

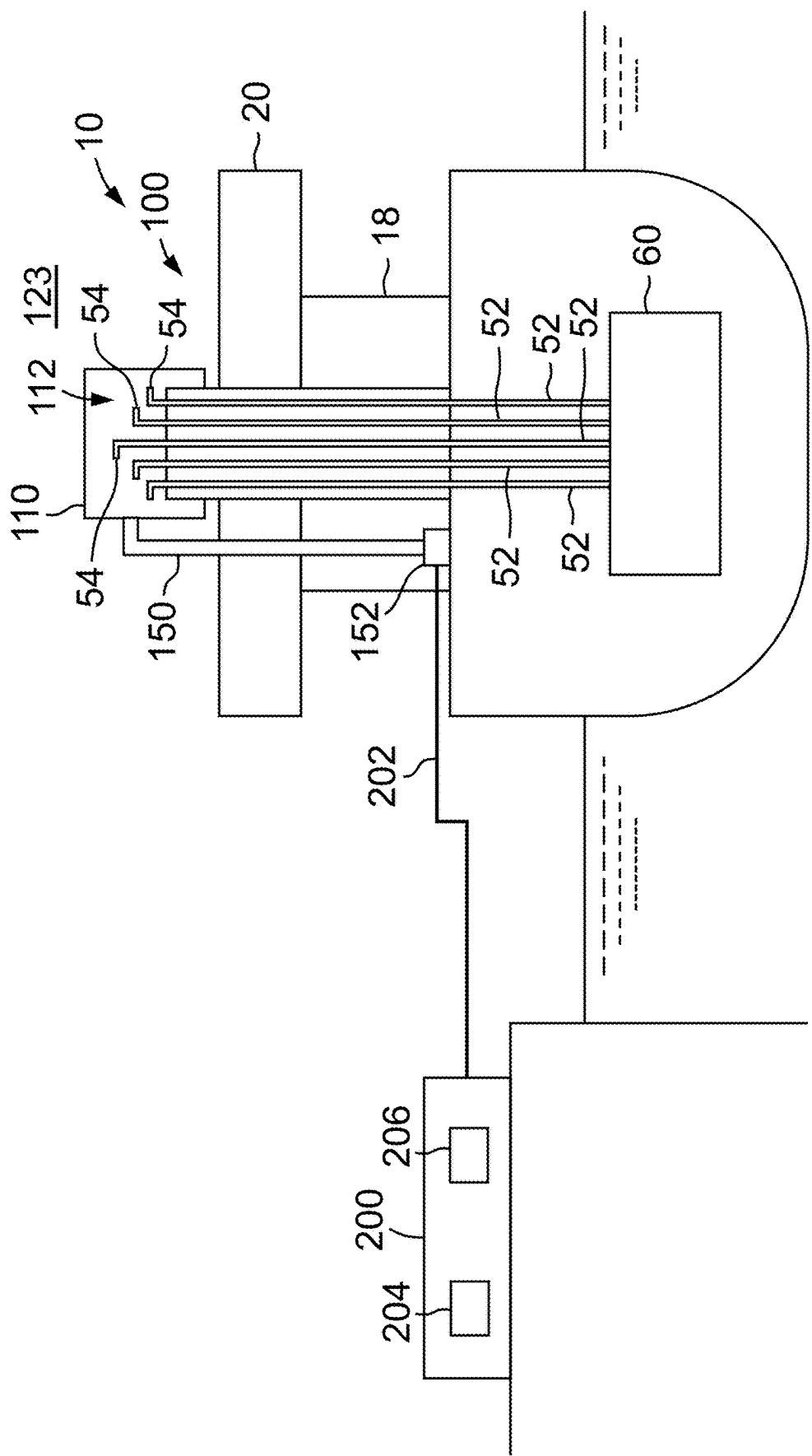


FIG. 8

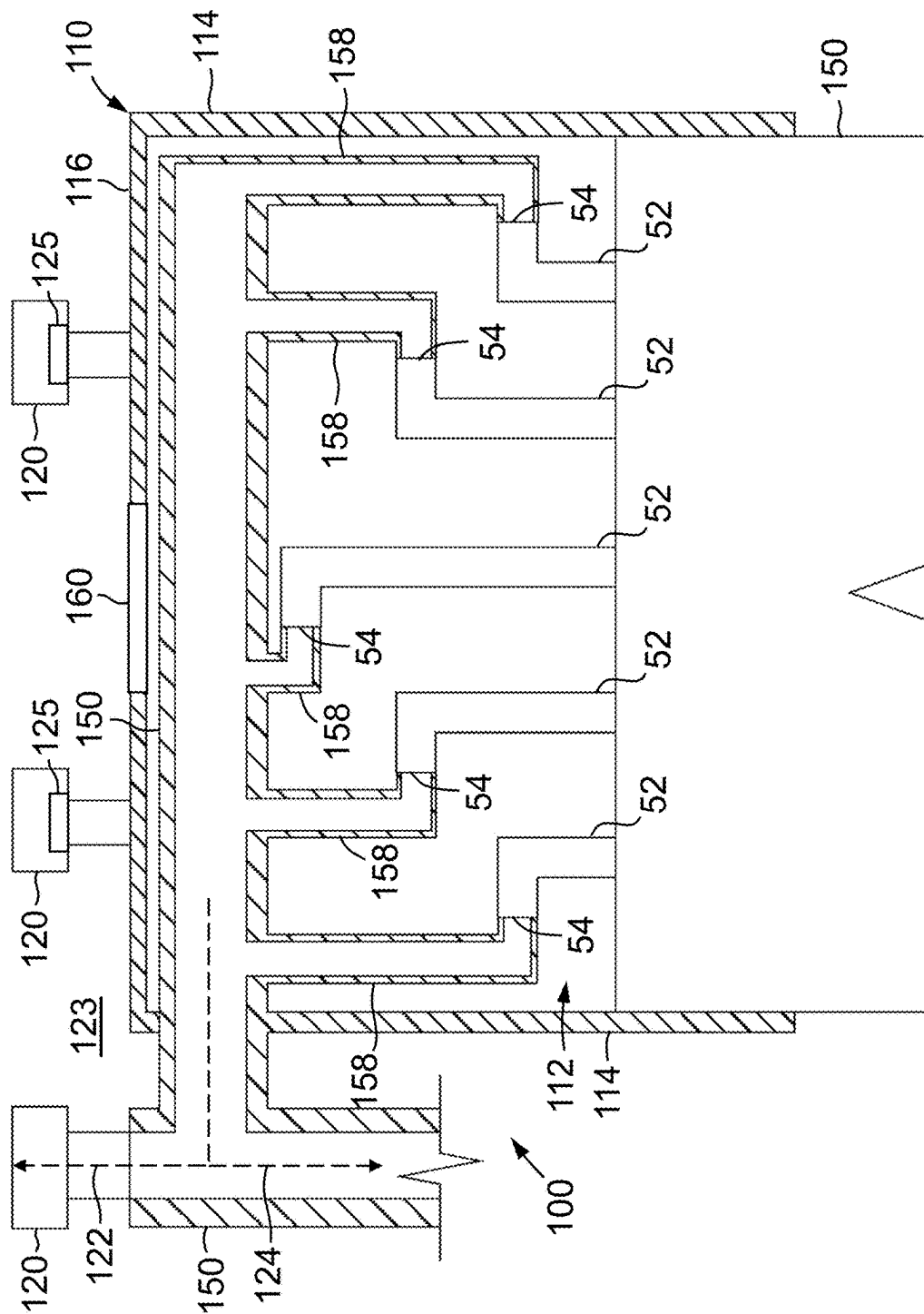


FIG. 9

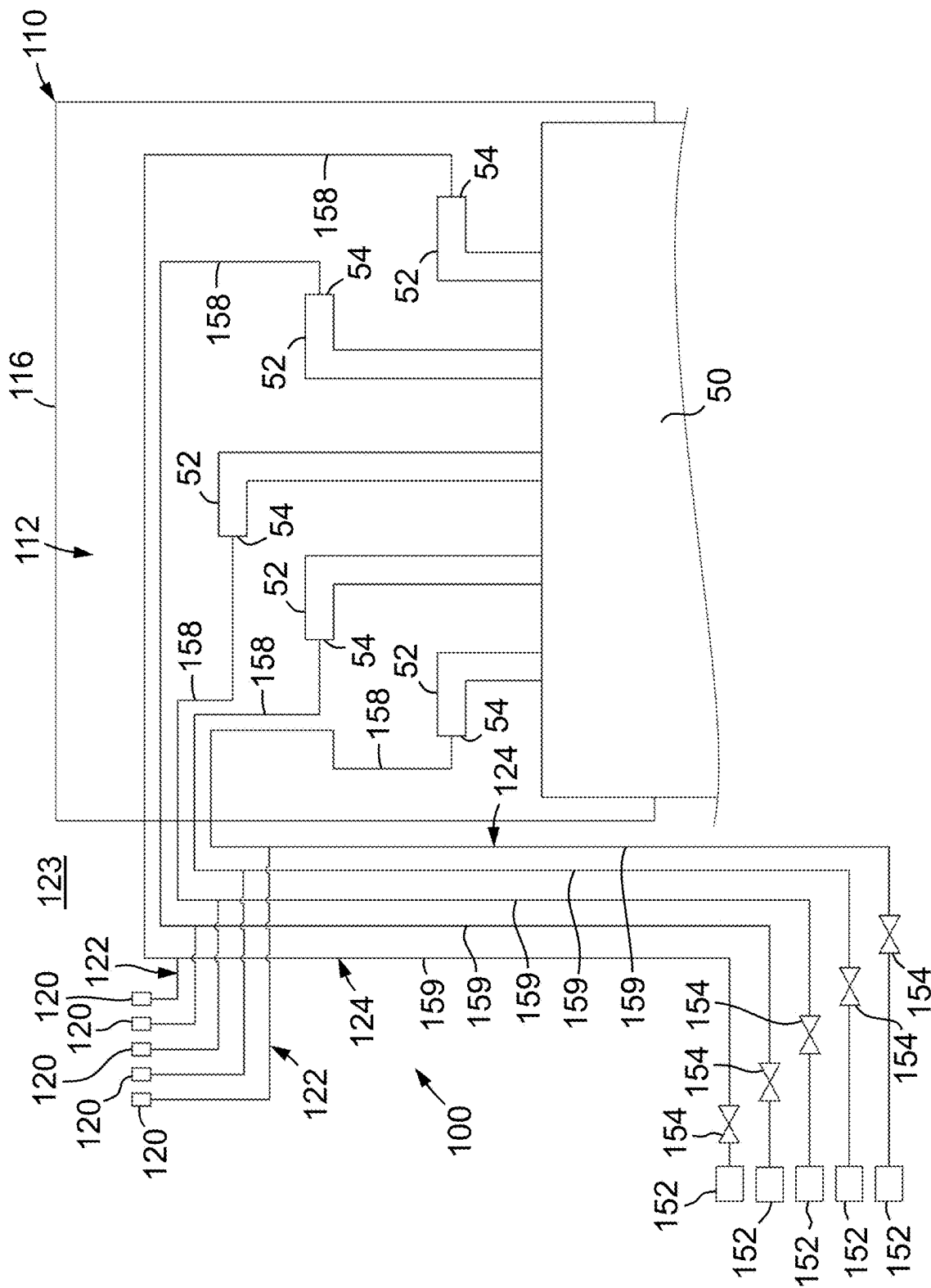


FIG. 10

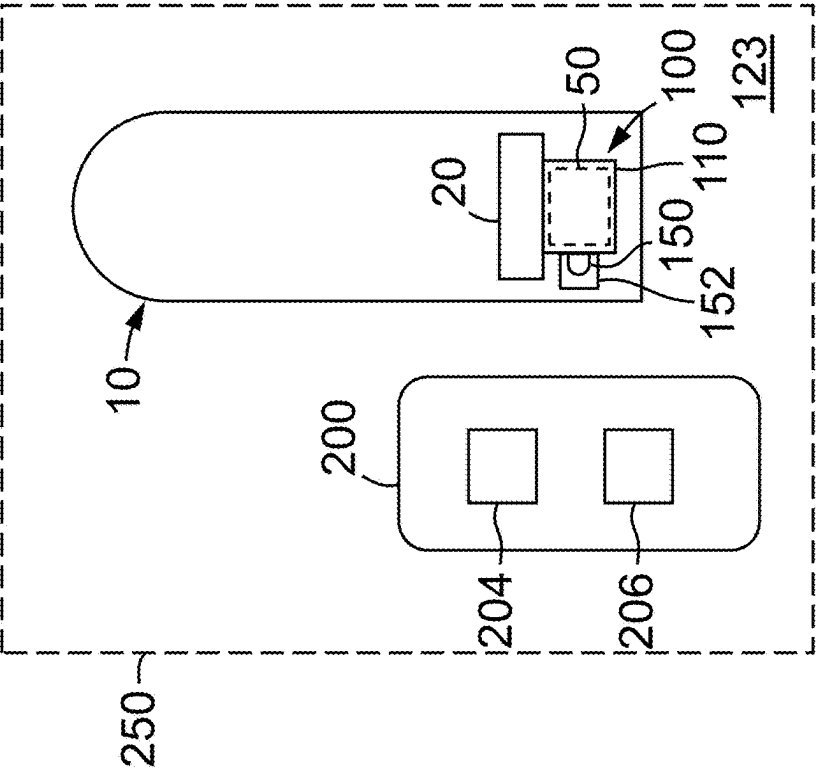


FIG. 11B

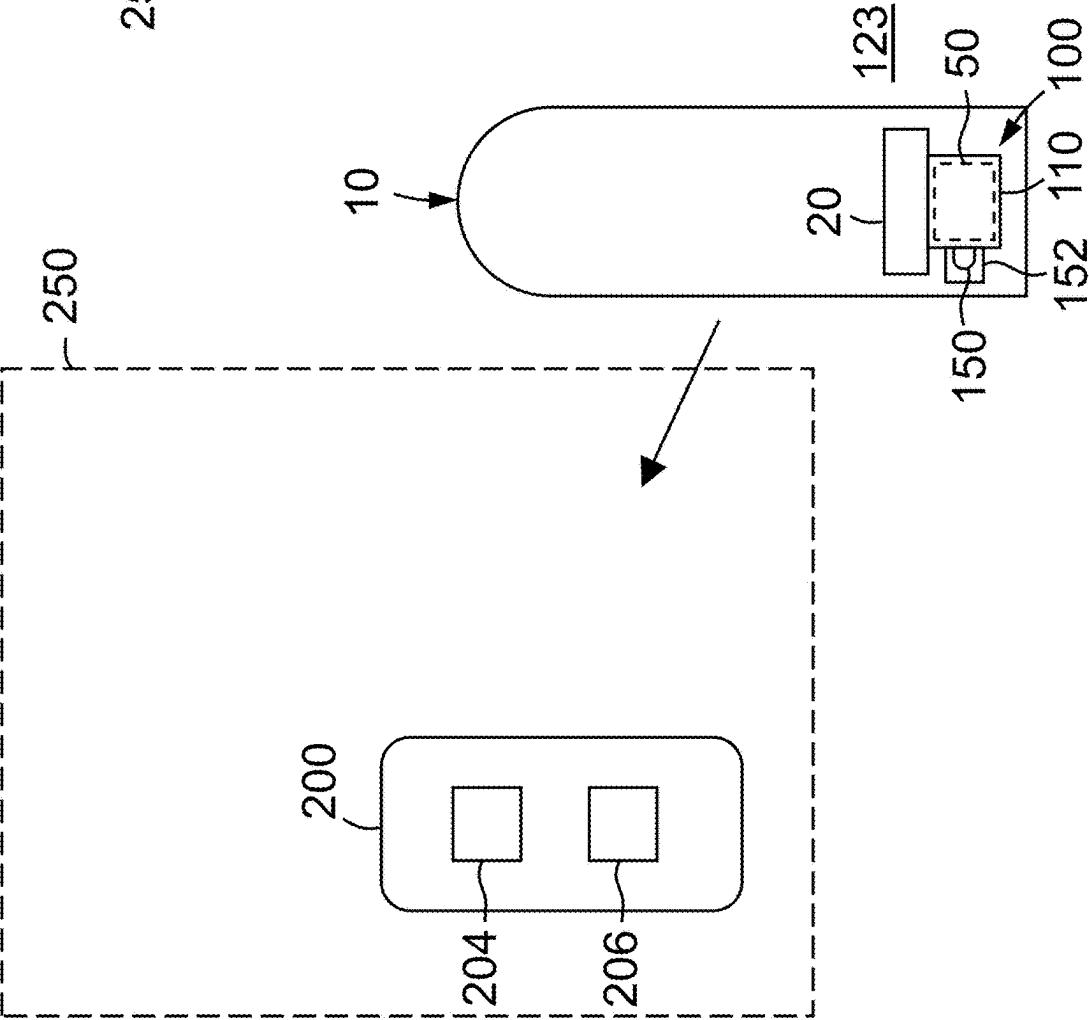


FIG. 11A

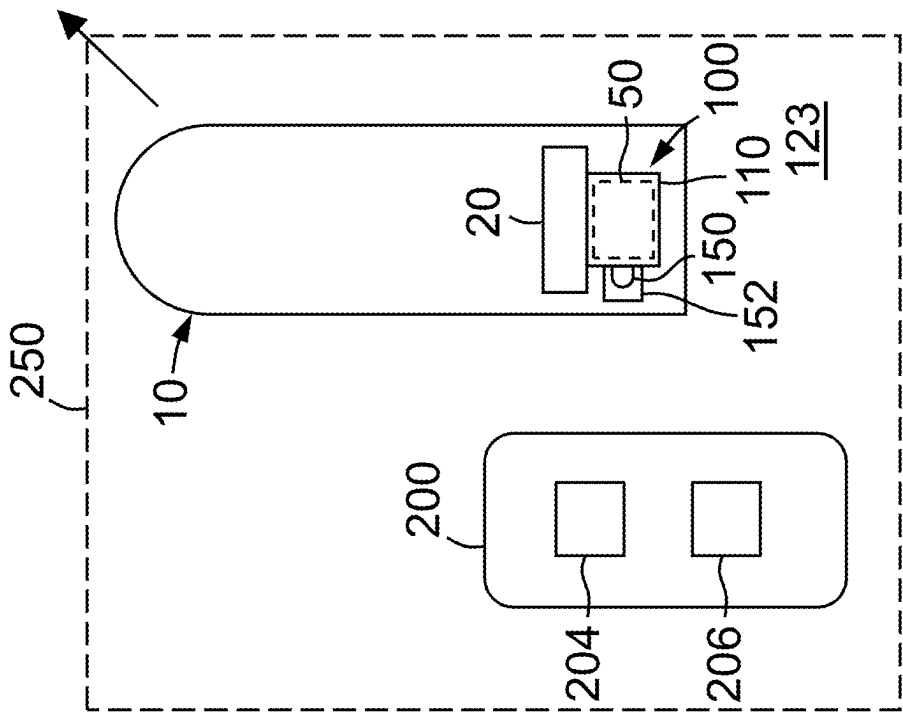


FIG. 11D

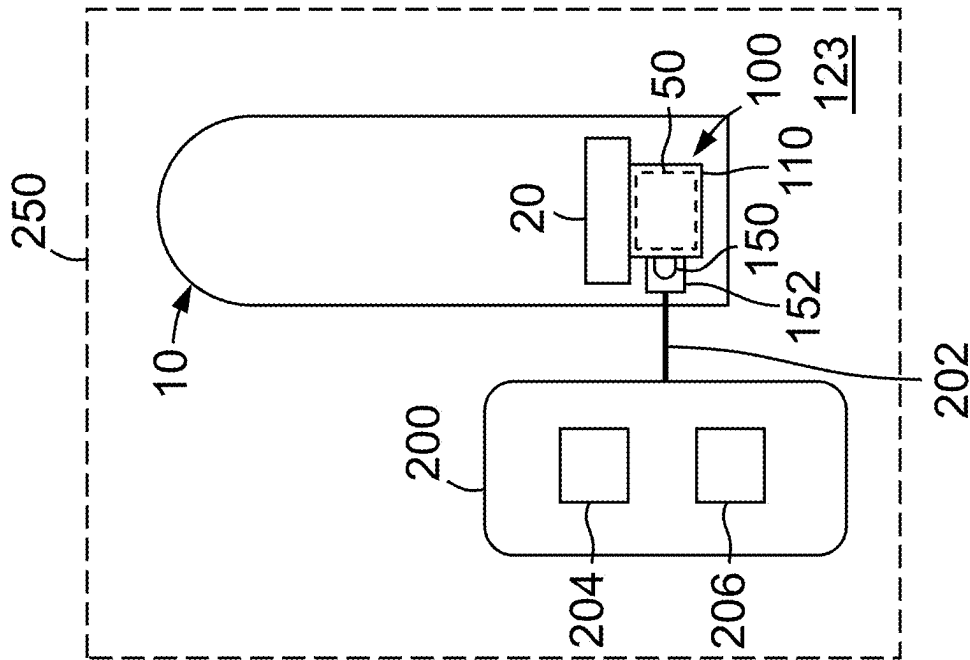


FIG. 11C

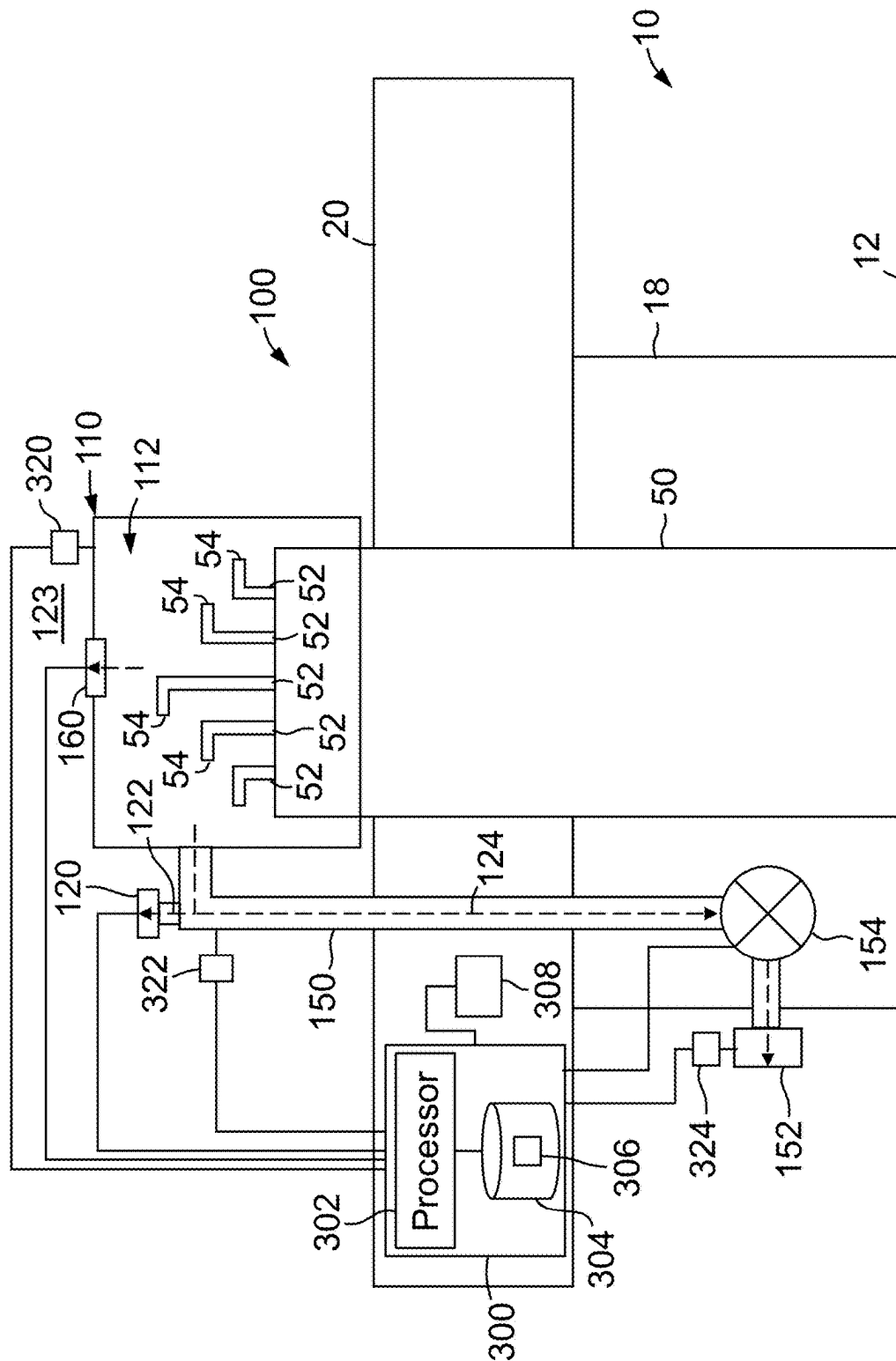
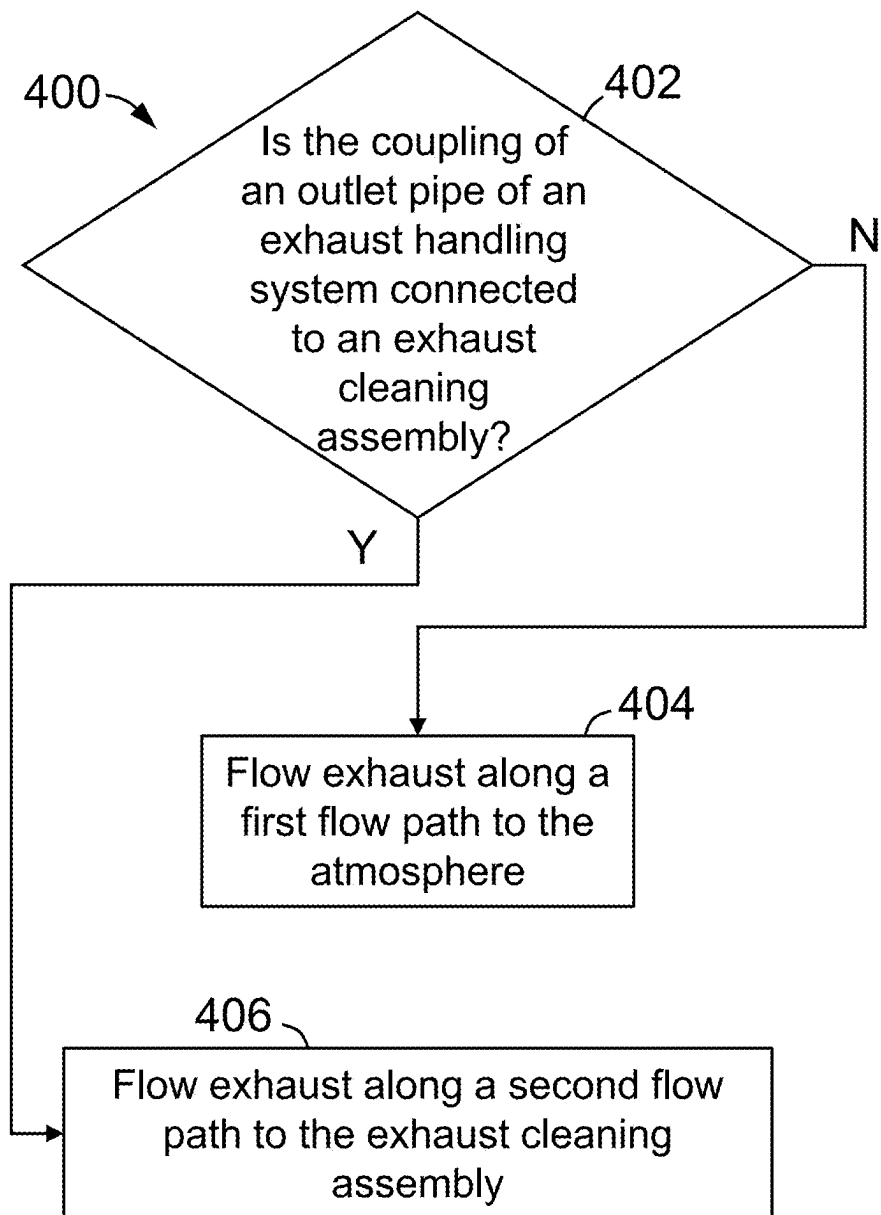
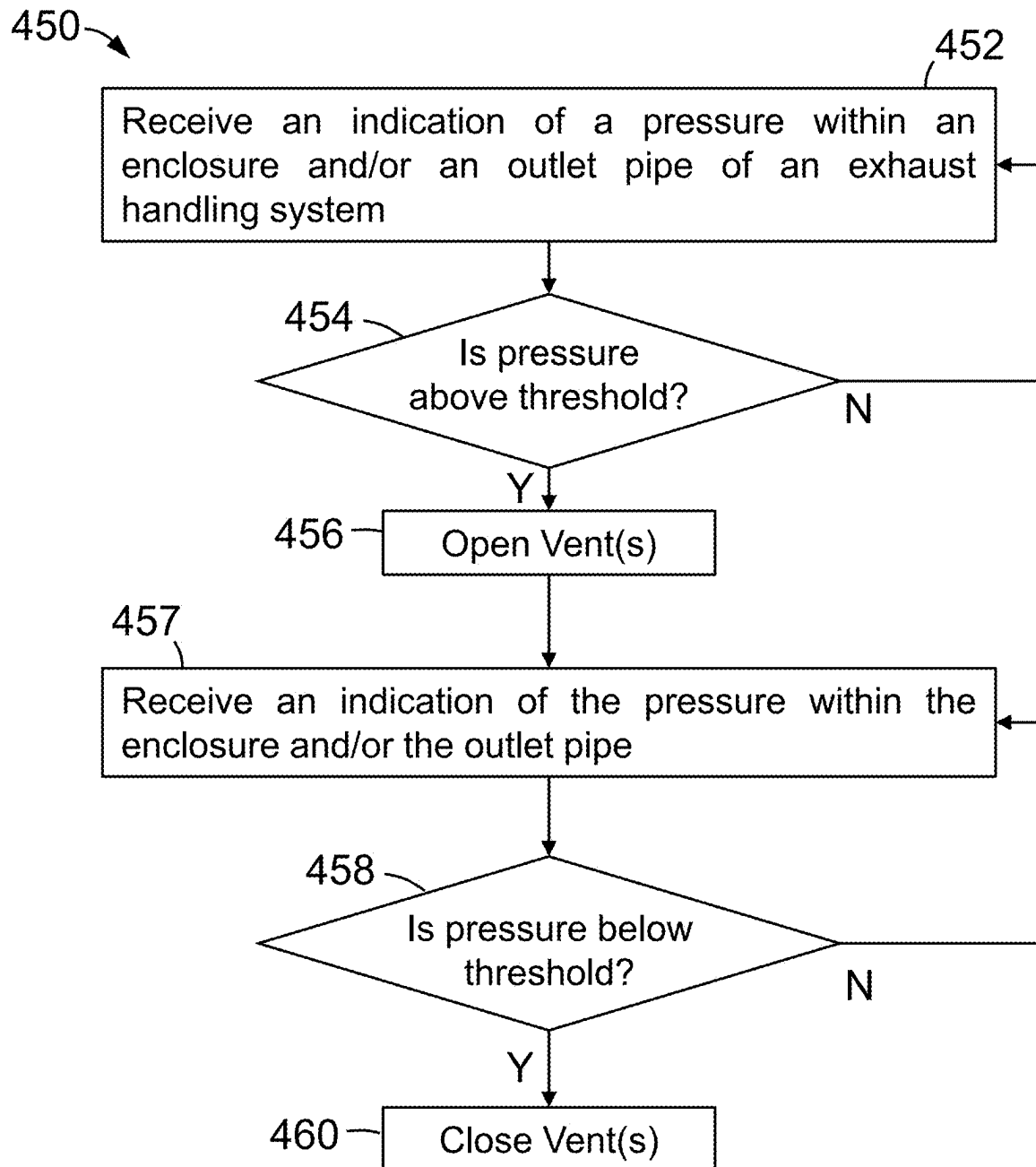


FIG. 12

**FIG. 13**

**FIG. 14**

1

EXHAUST HANDLING SYSTEMS FOR MARINE VESSELS AND RELATED METHODS

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. Non-Provisional application Ser. No. 18/369,485, filed Sep. 18, 2023, titled “Exhaust Handling Systems for Marine Vessels and Related Methods,” which claims priority to and the benefit of U.S. Provisional Application No. 63/485,886, filed Feb. 18, 2023, titled “Exhaust Handling Systems for Marine Vessels and Related Methods,” and U.S. Provisional Application Ser. No. 63/488,574, filed Mar. 6, 2023, titled “Exhaust Handling Systems for Marine Vessels and Related Methods,” the contents of which are incorporated herein by reference in their entirety.

BACKGROUND

This disclosure generally relates to marine vessels that travel on navigable bodies of water. More particularly, this disclosure relates to exhaust handling systems for marine vessels and related methods.

A marine vessel may include any suitable vessel or boat that is transportable or movable across a navigable body of water (such as an ocean, lake, river, etc.). Such marine vessels may include engines, motors, generators, and other systems configured to output exhaust fluids (or more simply “exhaust”). Typically, the exhaust is emitted to the atmosphere during operations. However, when multiple marine vessels are concentrated in the same geographical area, the combined exhaust from the marine vessels may significantly degrade the local air quality. Berthing locations such as ports, piers, harbors, moorings, etc., may represent locations where large numbers of marine vessels congregate. Thus, in a number of jurisdictions, rules or regulations regarding the output of exhaust at berthing locations have been or will be implemented to preserve the air quality both within the berthing location and in neighboring areas.

BRIEF SUMMARY

Some embodiments disclosed herein include exhaust handling systems for a marine vessel that are configured to allow selective collection of exhaust output from the exhaust-emitting systems of the marine vessel while at a berthing location. In some embodiments, the collected exhaust may be routed to an exhaust cleaning assembly that may store and/or process the exhaust to prevent harmful chemicals or pollutants within the exhaust from being emitted to the atmosphere. In some embodiments, the exhaust handling systems of the embodiments disclosed herein may include a cap that is connected to an upper end portion of an exhaust stack of the marine vessel so as to form an enclosure around one or more (such as a plurality of) exhaust pipes. The collected exhaust may be emitted directly to the atmosphere when the marine vessel is not at a berthing location (or during a situation that requires venting of exhaust to the atmosphere) or may be selectively routed to an exhaust cleaning assembly to avoid such atmospheric venting when the marine vessel is berthed. Accordingly, through use of the embodiments disclosed herein, the exhaust-emitting systems of a marine vessel may continue to operate when the marine vessel is berthed while preventing (or restricting) the emis-

2

sion of exhaust (or at least the harmful and/or polluting components thereof) to the atmosphere.

Some embodiments disclosed herein are directed to exhaust handling system for a marine vessel. In some embodiments, the exhaust handling system includes a cap connected to a top end portion of an exhaust stack of the marine vessel so as to form an enclosure that at least partially surrounds an outlet of an exhaust pipe extending through the exhaust stack. In addition, the exhaust handling system includes a collection pipe in fluid communication with the cap such that the collection pipe is configured to receive exhaust from the enclosure. Further, the exhaust handling system includes a coupling connected to the collection pipe that is configured to connect to an exhaust cleaning assembly. The exhaust cleaning system includes at least one tank to receive the exhaust. The cap at least partially defines a first flow path for the exhaust to flow from the enclosure to an atmosphere surrounding the cap. The collection pipe at least partially defines a second flow path for the exhaust to flow from the enclosure to the coupling via the collection pipe.

In some embodiments, the exhaust handling system includes a cap connected to a top end portion of an exhaust stack of the marine vessel so as to form an enclosure that at least partially surrounds an outlet of an exhaust pipe extending through the exhaust stack. In addition, the exhaust handling system includes a vent in fluid communication with the enclosure, the vent including at least one valve member that is actuatable between a first position to emit exhaust from the enclosure to an atmosphere surrounding the cap via the vent and a second position to prevent an emission of exhaust from the enclosure to the atmosphere via the vent. Further, the exhaust handling system includes a collection pipe in fluid communication with the enclosure, and a coupling connected to the collection pipe and configured to connect to an exhaust cleaning assembly that includes at least one tank to receive the exhaust.

Some embodiments disclosed herein are directed to exhaust handling system for a marine vessel. In some embodiments, the exhaust handling system includes a cap connected to a top end portion of an exhaust stack of the marine vessel such that the cap is supported by the exhaust stack and such that the cap forms an enclosure that at least partially surrounds an outlet of an exhaust pipes extending through the exhaust stack. In addition, the exhaust handling system includes a pressure-actuated vent in fluid communication with the enclosure. Further, the exhaust handling system includes a collection pipe in fluid communication with the enclosure and configured to connect to an exhaust cleaning assembly that includes at least one tank to receive the exhaust.

Some embodiments are directed to methods. In some embodiment, the method includes (a) positioning a marine vessel in a berthing location, the marine vessel including a deck, an exhaust stack having an top end portion positioned above from the deck, an exhaust pipe extending through the exhaust stack to an outlet that is positioned above the top end portion of the exhaust stack, a cap connected to the exhaust stack so as to form an enclosure that at least partially surrounds the outlet of the exhaust pipe, and an collection pipe in fluid communication with the enclosure. In addition, the method includes (b) connecting an exhaust cleaning assembly to the collection pipe after (a), the exhaust cleaning assembly including at least one tank to receive the exhaust. Further, the method includes (c) operating the

3

marine vessel to flow an exhaust emitted from the outlet of the exhaust pipe to the exhaust cleaning assembly after (b) via the collection pipe.

In some embodiments, the method includes (a) positioning a marine vessel in a berthing location, the marine vessel including an exhaust stack, an exhaust pipe extending through the exhaust stack to an outlet that is positioned above a top end portion of the exhaust stack, a cap that is connected to the exhaust stack to form an enclosure that surrounds the outlet of the exhaust pipe. In addition, the method includes (b) flowing an exhaust from the enclosure to an atmosphere surrounding the cap via a first flow path during (a), the first flow path at least partially defined by the cap. Further, the method includes (c) connecting an exhaust cleaning assembly positioned at the berthing location to the cap after (a), the exhaust cleaning system including at least one tank to receive the exhaust. Still further, the method includes (d) flowing the exhaust from the enclosure to the exhaust cleaning assembly via a second flow path after (b), the second flow path at least partially defined by the cap.

Embodiments described herein include a combination of features and characteristics intended to address various shortcomings associated with certain prior devices, systems, and methods. The foregoing has outlined rather broadly the features and technical characteristics of some of the disclosed embodiments in order that the detailed description that follows may be better understood. The various characteristics and features described above, as well as others, will be readily apparent to those having ordinary skill in the art upon reading the following detailed description, and by referring to the accompanying drawings. It should be appreciated that this disclosure may be readily utilized as a basis for modifying or designing other structures for carrying out the same purposes as the disclosed embodiments. It should also be realized that such equivalent constructions do not depart from the spirit and scope of the principles disclosed herein.

BRIEF DESCRIPTION OF THE DRAWINGS

For a detailed description of various embodiments, reference will now be made to the accompanying drawings in which:

FIG. 1 is a schematic side view of a marine vessel having an exhaust handling system according to some embodiments of the disclosure;

FIG. 2 is a schematic rear view of the marine vessel of FIG. 1 according to some embodiments of the disclosure;

FIG. 3 is a perspective view of a cap of the exhaust handling system of the marine vessel of FIG. 1 according to some embodiments of the disclosure;

FIG. 4 is a cross-sectional view taken along section A-A in FIG. 3 according to some embodiments of the disclosure;

FIGS. 5 and 6 are side, cross-sectional views of a vent for a cap of the exhaust handling system of FIG. 3 according to some embodiments of the disclosure;

FIG. 7 is a schematic view of the marine vessel of FIG. 1 connected to an exhaust cleaning assembly positioned on a barge according to some embodiments of the disclosure;

FIG. 8 is a schematic view of the marine vessel of FIG. 1 connected to an exhaust cleaning assembly positioned on-shore according to some embodiments of the disclosure;

FIG. 9 is a cross-sectional view taken along section A-A in FIG. 3 according to some embodiments of the disclosure;

FIG. 10 is a schematic view of the cap of the exhaust handling system of the marine vessel of FIG. 1 according to some embodiments of the disclosure;

4

FIGS. 11A-11D are sequential top views of a sequence or method for routing exhaust from the marine vessel of FIG. 1 via the exhaust handling system when the marine vessel is berthed at a berthing location and when the marine vessel is traveling into and out of the berthing location according to some embodiments of the disclosure;

FIG. 12 is a schematic view of the exhaust handling system of the marine vessel of FIG. 1 further showing a controller for controlling the flow of exhaust from the marine vessel via the exhaust handling system according to some embodiments of the disclosure; and

FIGS. 13 and 14 are diagrams of methods for flowing the exhaust of a marine vessel through an exhaust handling system according to some embodiments of the disclosure.

DETAILED DESCRIPTION

As previously described, the exhaust from marine vessels at a berthing location may degrade air quality and may even be restricted by local rules and regulations. However, ceasing operation of all exhaust-emitting systems on a marine vessel may not be desirable or feasible while the marine vessel is at the berthing location. For instance, electrical generation systems (for example, diesel generators, turbine generators, etc.) may continue to operate so that other electrically operated systems and assemblies of the marine vessel (for example, communications systems, safety systems, control systems, water and sewage systems, HVAC systems, etc.) may also be operated while at the berthing location. In addition, some berthing locations do not have sufficient infrastructure to support and operate these various systems and assemblies of the berthed marine vessels independently of the onboard exhaust-emitting systems.

Accordingly, embodiments disclosed herein include exhaust handling systems for a marine vessel that are configured to allow selective collection of exhaust output from the exhaust-emitting systems of the marine vessel while at a berthing location. In some embodiments, the collected exhaust may be routed to an exhaust cleaning assembly that may store and/or process the exhaust to prevent harmful chemicals or pollutants within the exhaust from being emitted to the atmosphere. In some embodiments, the exhaust handling systems of the embodiments disclosed herein may include a cap that is fixed to an upper end of an exhaust stack of the marine vessel so as to form an enclosure around one or more (e.g., a plurality of) exhaust pipes. The collected exhaust may be emitted directly to the atmosphere when the marine vessel is not at a berthing location (or during a situation that requires venting of exhaust to the atmosphere) or may be selectively routed to an exhaust cleaning assembly to avoid such atmospheric venting when the marine vessel is berthed. Accordingly, through use of the embodiments disclosed herein, the exhaust-emitting systems of a marine vessel may continue to operate when the marine vessel is berthed while preventing (or restricting) the emission of exhaust (or at least the harmful and/or polluting components thereof) to the atmosphere.

Reference is now made to FIG. 1, which shows a marine vessel 10 and an exhaust handling system 100 according to some embodiments. The marine vessel 10 may include any suitable vessel or ship that may travel within or across a navigable body of water such as, for instance, an ocean, a sea, a lake, a river, a channel, etc. In some embodiments, the marine vessel 10 may be a cargo vessel, such as a container ship, tankship, reefer ship, etc.

5

Marine vessel **10** includes a rear end (or stern) **10a**, a front end (or bow) **10b** opposite rear end **10a**, and a main deck (or more simply “deck”) **12** extending between ends **10a**, **10b**. Deck **12** may define an exterior surface (or collection of exterior surfaces) on the marine vessel **10** that may be accessed by personnel.

A rudder **14** and propeller **16** may be positioned at (or proximate to) rear end **10a**. As is known to one having ordinary skill in the art, the propeller **16** may provide propulsion to the marine vessel **10**, and the rudder **14** may be turned to steer the marine vessel **10** within a body of water.

An accommodation deck (or more simply “accommodation”) **18** is positioned atop deck **12**. The accommodation **18** is the living space of the marine vessel **10** and may include one or more cabins (or rooms), galleys, store rooms, mess-rooms, or other rooms that may be used by personnel. A bridge **20** may be positioned atop (or adjacent to) the accommodation **18**. The bridge **20** may include one or more controls for the marine vessel **10** including (for instance) steering controls, communications systems, other system controls, etc.

As shown in FIGS. 1 and 2, an exhaust stack **50** may extend upward and away from deck **12**. The exhaust stack **50** may sometimes be referred to as an “exhaust funnel.” The exhaust stack **50** may be positioned adjacent the accommodation **18** and bridge **20** along deck **12**. Specifically, the exhaust stack **50** may be positioned between bridge **20** and rear end **10a** along deck **12** in some embodiments. Exhaust stack **50** may extend upward from deck **12** to a top end portion **50a**. One or more (such as one or a plurality of) exhaust pipes **52** extend through exhaust stack **50** and out of the top end portion **50a**. Specifically, each exhaust pipe **52** has an outlet **54** that is extended out of the exhaust stack **50** and positioned above the top end portion **50a**.

During operations, each exhaust pipe **52** (or some of the exhaust pipes **52**) may output exhaust fluid (or “exhaust”). Specifically, as shown in FIG. 2, each exhaust pipe **52** is fluidly connected to one or more exhaust-emitting systems **62**, **64**, **66** (collectively referred to as “exhaust-emitting systems **60**”). For instance, the exhaust-emitting systems **60** of marine vessel **10** may include diesel electric generator(s) **62**, boilers **64**, inert gas systems **66**, or other systems or assemblies that may output combustion (or other) exhaust during operations. Accordingly, marine vessel **10** includes an exhaust handling system **100** that is configured to selectively collect and route exhaust emitted from the exhaust pipes **52** to an exhaust cleaning assembly (not shown in FIG. 1 but see exhaust cleaning assembly **200** shown in FIGS. 7 and 8 and described herein).

As shown in FIGS. 1 and 2, exhaust cleaning assembly **100** includes a cap **110** that is connected to exhaust stack **50**. Specifically, cap **110** may be connected to top end portion **50a** of exhaust stack **50** so as to form or define an enclosure **112** that surrounds (or at least partially surrounds) the outlets **54** of each of the exhaust pipes **52**. As will be described in more detail below, the enclosure **112** may collect exhaust emitted from the outlets **54** of exhaust pipes **52**. A collection pipe **150** extends from cap **110** toward the deck **12** and is connected to the cap **110** such that the collection pipe **150** is in fluid communication with the enclosure **112**. Thus, during operations, exhaust may be collected within enclosure **112** and routed toward deck **12** via collection pipe **150**.

In some embodiments, the cap **110** may be connected to the exhaust stack **50** so that the outlets **54** of less than all of the exhaust pipes **52** are surrounded (at least partially) by the enclosure **112**. Thus, in some embodiments, the outlets **54** of

6

one or more of the exhaust pipes **52** may be positioned outside of the enclosure **112** and the outlets of one or more of the exhaust pipes **52** may be positioned inside the enclosure **112**. For instance, without being limited to this or any other theory, outlet pipes **52** associated with the main engine of the marine vessel **10** may not normally emit exhaust (or may not emit a substantial volume of exhaust) when the marine vessel **10** is stationary at a berthing location. Thus, the outlet pipes **52** associated with the main engine of the marine vessel **10** may not be positioned in the enclosure **112** so as to minimize a size and complexity of the cap **110** as well as to avoid constriction of the exhaust flow out from the main engine of the marine vessel **10** when the marine vessel **10** is not at a berthing location. In some embodiments, multiple caps **110** may be attached to exhaust stack **50** (or to multiple exhaust stacks **50** depending on the configuration of the marine vessel **10**) so that different caps **110** may surrounding (at least partially) different ones or groups of the exhaust pipes **52** during operations. For instance, in some embodiments, a first cap **110** may be positioned on a first exhaust stack **50** so as to surround (at least partially) one or more exhaust pipes **52** extending therethrough, and a second cap **110** may be positioned on a second exhaust stack **50** so as to surround (at least partially) one or more exhaust pipes **52** extending therethrough.

Collection pipe **150** (or a portion of collection pipe **150**) may include a rigid pipe that is rigidly connected (such as via brackets, welding, bolting, riveting, etc.) to cap **110** and marine vessel **10** (FIGS. 1 and 2). In some embodiments, the collection pipe **150** (or a portion of the collection pipe **150**) may include flexible ductwork (or ducting) such as, for instance, flexible hoses and the like that may be readily deformed and maneuvered during operations. In some embodiments, the collection pipe **150** may be temporarily connected to the cap **110** and/or the marine vessel **10** so as to allow collection pipe **150** to be easily removed when not in use (such as when marine vessel **10** is not at a berthing location). In addition, in some embodiments, the collection pipe **150** (whether it includes rigid pipe and/or flexible ductwork) may include one or more layers of thermal insulation.

Collection pipe **150** may include or be connected to a coupling **152** that, as will be described in more detail below, may be connected to an exhaust cleaning assembly during operations. The coupling **152** may be positioned at or proximate to the deck **12** such that the coupling **152** may be accessible from the deck **12**. Thus, during operations, personnel may interact with the coupling **152** (for example, to connect the coupling **152** and collection pipe **150** to an exhaust cleaning assembly) from the deck **12**. Further details of the exhaust handling system **100** are now described below according to some embodiments.

As shown in FIGS. 3 and 4, in some embodiments the cap **110** may include an upper closed end **110a** (or more simply “upper end **110a**”) and a lower open end **110b** (or more simply “lower end **110b**”) opposite the upper end **110a**. In some embodiments, the cap **110** may be shaped as a rectangular parallelepiped, such that the cap **110** includes a planar top **116** and a plurality of (e.g., four) planar sides **114** extending from planar top **116** to lower end **110b**. However, other shapes are contemplated for cap **110**, such as, for instance, cylindrical, conical, triangular prism, irregular shape, etc. As best shown in FIG. 4, the top **116** and sides **114** define the enclosure **112**. During operations, cap **110** may be connected to top end portion **50a** of exhaust stack **50** such that the enclosure **112** surrounds the outlets **54** of the exhaust pipes **52** as previously described. In addition, the

collection pipe 150 may be fluidly connected to the enclosure 112 on one of the planar sides 114.

A plurality of flow paths are defined within exhaust handling system 100 to route exhaust out of the enclosure 112 of cap 110 during operation. For instance, as shown in FIG. 4, one or more first flow paths 122 may allow exhaust to flow from the enclosure 112 to the atmosphere 123 surrounding the cap 110. For instance, the first flow path(s) 122 may extend out of the enclosure 112 and through one or more vents 120, 160 to the surrounding atmosphere 123. Thus, the first flow path(s) 122 may be at least partially defined by the cap 110 and one or more of the vents 120, 160.

The vents 120 are connected to the cap 110 such that they are in fluid communication with the enclosure 112. The vents 120 may include an actuatable vent. For example, in some embodiments, the vents 120 may have a valve or valve member 125 (such as a gate valve, flapper valve, butterfly valve, etc.) that is actuatable between an open position and a closed position. When the valve member 125 of a vent 120 is in the open position, the vent 120 may allow exhaust to flow therethrough and into the surrounding atmosphere 123, and when the valve member 125 of a vent 120 is in the closed position, the vent 120 may prevent (or restrict) the flow of exhaust therethrough to the surrounding atmosphere 123. In some embodiments, the valve members 125 of vents 120 may be actuated between the open and closed positions by a controller (such as controller 300 shown in FIG. 12 and described herein) and/or manually by personnel. In some embodiments, the vents 120 (more specifically the valve members 125) may be pressure actuated (such that the vents 120 may be “pressure-actuated vents”). Specifically, in some embodiments, the valve member 125 of one or more of the vents 120 may be biased such that the valve member 125 is configured to open when a sufficient differential pressure is applied across the vent 120 (and against the bias applied to the valve member 125). For instance, in some embodiments, the valve member(s) 125 of one or more of the vents 120 may be biased to the closed position (for example, via a spring or other suitable biasing member or assembly), and when a pressure within the enclosure 112 rises above a threshold, the valve member(s) 125 of the one or more of the vents 120 may transition from the closed position to the open position to allow exhaust to flow out of the enclosure 112 to the surrounding atmosphere 123.

The vent 160 may be connected to cap 110 such that vent 160 is in fluid communication with the enclosure. The vent 160 may be positioned along the planar top 116 of cap 110 and may be also be configured to transition between a closed position and an open position to selectively prevent and allow exhaust to flow out of enclosure 112 to the surrounding atmosphere 123, respectively. As shown in FIG. 5, in some embodiments, the vent 160 may include a housing 162 that defines an opening or flow path 164 therethrough. The housing 162 is mounted to the planar top 116 so that the opening 164 forms or defines an opening in the planar top 116 of cap 110. A plurality of louvers 166 are pivotably connected to the housing 162 such that the louvers 166 are parallel to one another and span across the opening 164. Each louver 166 is pivotably connected to the housing 162 via a corresponding hinge 168 so that the louvers 166 may pivot about the hinge 168, relative to the housing 162 during operations. Specifically, the louvers 166 may be pivoted about the hinges 168 between a first or closed position shown in FIG. 5 and a second or open position shown in FIG. 6 (as well as a plurality of positions between the closed position of FIG. 5 and the open position of FIG. 6). Thus, the

vent 160 may be referred to herein as an “actuatable vent” similar to embodiments of the vent 120 previously described, and the louvers 166 may be referred to as actuatable “valve members” or “valves” similar to embodiments of the valve members 125 of vents 120 previously described.

When the louvers 166 are in the closed position (FIG. 5), the louvers 166 may engage, interlock, overlay, overlap, and/or otherwise cooperate with one another so as to cover and therefore occlude the opening 164. As a result, when the louvers 166 are in the closed position (FIG. 5), exhaust is prevented (or is at least restricted) from flowing out of the enclosure 112 to the surrounding atmosphere 123. In some embodiments, the louvers 166 may include or be connected to seals (such as compliant seals) that may further enhance the ability of louvers 166 to prevent (or at least restrict) the flow of exhaust out of the enclosure 112 via the opening 164 when louvers 166 are in the closed position (FIG. 5). Conversely, when the louvers 166 are in the opening position (FIG. 6), the louvers 166 may rotate or pivot about the hinges 168 so as to disengage with one another and thereby open or uncover the opening 164 of housing 162 and place the surrounding atmosphere 123 in fluid communication with the enclosure 112.

The louvers 166 may be synchronously rotated about the hinges 168 between the closed position (FIG. 5) and the open position (FIG. 6) by a suitable transmission or connection 163 (such as a connection bar, gears-such as rack and pinion gears, or other suitable connection device or assembly). A driver 170 may be connected to the louvers 166 (such as directly connected to louvers 166 or indirectly connected to louvers 166 via connection 163). In some embodiments, the driver 170 may include a suitable motor, such as, for instance, an electric motor, hydraulic motor, pneumatic motor, etc. In some embodiments, the driver 170 may include a manual driver such as a chain pull, lever, or other manually operated device. During operations, the driver 170 may actuate the louvers 166 (such as via the connection 163) to rotate or transition between the closed position (FIG. 5) and the closed position (FIG. 6) to selectively prevent (or restrict) or allow, respectively, exhaust to flow out of the enclosure 112 to the surrounding atmosphere 123. In some embodiments, the vent 160 may also include a fan or blower that may draw exhaust out of the enclosure 112 and through the opening 164 and into the atmosphere 123 when the louvers 166 are in the open position (FIG. 6). Still further, in some embodiments, the louvers 166 may be actuated between the open position and the closed position by a differential pressure applied across the vent 160 between the enclosure 112 and the atmosphere 123. For instance, the driver 170 may be connected to or may include a pressure sensor and may actuate the louvers 166 based on an output thereof. As another example, the louvers 166 may be rotationally biased (for example, via torsional springs or other biasing members or systems) toward the closed position (FIG. 5) and may transition to the open position (FIG. 6) in response to a sufficient differential pressure between the enclosure 112 and atmosphere 123. Thus, the vent 160 may also be referred to herein as a “pressure-actuated vent” as described herein.

As shown in FIG. 4, in some embodiments, first flow path(s) 122 may extend through one or more of the vents 120, 160 positioned on the planar top 116 of the cap 110. In some embodiments, the first flow path(s) 122 may extend through vent(s) 120 and/or vents 160 that are positioned along collection pipe 150. In some embodiments, a plurality of first flow paths 122 may extend out of the enclosure 112

via vents **120**, **160** in the planar top **116** and/or the collection pipe **150**; however, in some embodiments, a single first flow path **122** may extend out of the enclosure **112** via a vent **120** or a vent **160** in either the planar top **116** or the collection pipe **150**. Still further, in some embodiments, a first flow path **122** may extend out of the enclosure **112** via a vent **120** and/or a vent **160** that is positioned along and/or coupled to one of the planar sides **114** of cap **110**.

As shown in FIG. 4, one or more second flow paths **124** may allow exhaust to flow out of the enclosure **112** via the collection pipe **150**. Thus, the one or more second flow paths **124** may be at least partially defined by the cap **110** and the collection pipe **150**. As previously described, the collection pipe **150** may extend from the cap **110** to the coupling **152** (that may be accessible from the deck **12** of marine vessel **10** as previously described). In particular, the coupling **152** may include any suitable coupling mechanism that is configured to connect the collection pipe **150** to an exhaust cleaning assembly (such as the exhaust cleaning assembly **200** shown in FIGS. 7 and 8) via a suitable conduit (for example, a hose, pipe, tubing, etc.). For instance, in some embodiments, the coupling **152** may include a flanged coupling, quick connect coupling, threaded coupling, union, clamped coupling, or some combination thereof. A valve **154** is positioned along the collection pipe **150** between the enclosure **112** and the coupling **152**. The valve **154** may be actuated between an open position to allow exhaust to flow along collection pipe **150** via second flow path **124** to coupling **152**, and a closed position to prevent (or at least restrict) the flow of exhaust along the collection pipe **150** via second flow path **124** to coupling **152**. In some embodiments, the valve **154** may be actuated between the open and closed positions by a controller (such as the controller **300** shown in FIG. 12) and/or manually by personnel.

A cleanout port **156** may be positioned along the collection pipe **150** between the coupling **152** and the enclosure **112**. Specifically, the cleanout port **156** may be positioned along collection pipe **150** so that it is accessible from the deck **12** of marine vessel **10**. The cleanout port **156** is configured to provide access into the collection pipe **150** independent of the coupling **152** so that personnel may clean out or remove debris that may collect within the collection pipe **150** during operations and thereby prevent the second flow path **124** (or one or more of the first flow paths **122**) from becoming obstructed. In some embodiments, the clean out port **156** may be closed or occluded via a flanged cap **157**; however, any suitable capping or closing device (such as a hatch, blind, etc.) may be utilized to close the cleanout port **156** in some embodiments.

In some embodiments, the cap **110** may be permanently or fixedly installed on the marine vessel **10**. Specifically, the cap **110** may remain connected and fixed to the exhaust stack **50** both when the marine vessel is at berth (such as at a berthing location) and when the marine vessel **10** is not at a berthing location and is moving across a body of water (for example, across an ocean or lake). Thus, the cap **110** may remain fixed to the exhaust stack **50** so that the cap **110** may not be readily lifted or removed from the exhaust **50** without breaking or disconnecting the connections between the cap **110** and exhaust stack **50**. In some embodiments, the cap **110** may be welded to the exhaust stack **50**. In some embodiments, the cap **110** may be integrally formed as part of the exhaust stack **50** itself. In some embodiments, the cap **110** may be fixed to the exhaust stack **50** via one or more of a bolted connection, a riveted connection, or other suitable connections or structures.

In some embodiments, the cap **110** may be temporarily connected to the exhaust stack **50** such that the cap **110** may be installed on the exhaust stack **50** when the marine vessel **10** is positioned at a berthing location and then may be removed when the marine vessel **10** is to move out of (or away from) the berthing location. In some embodiments, the cap **110** may be temporarily installed on the exhaust stack **50** via clamps, a shouldered engagement (such as by resting the cap **110** on an external shelf or shoulder of the exhaust stack **50**), and/or any other suitable temporary connection.

Regardless as to whether the cap **110** is permanently or temporarily connected to the exhaust stack **50**, in some embodiments, the cap **110** may be supported (such as fully supported) by the exhaust stack **50**. That is, the weight of the cap **110** may be borne by the exhaust stack **50** (or other frames, structures, or other components of the marine vessel **10**) during operations so that a crane or other lifting device may not be attached to the cap **110** during operations. A crane (or other lifting device) may be utilized to lower the cap **110** onto exhaust stack **50** or lift the cap **110** away from the exhaust stack **50**; however, once the cap **110** is connected to the exhaust stack **50**, the crane (or other lifting device) may be disconnected from cap **110** and the weight of the cap **110** may be borne by the exhaust stack **50** and/or other portions or components of the marine vessel **10**.

The flow paths **122**, **124** may selectively route exhaust out of the enclosure **112** when the marine vessel **10** is and is not at berth. For instance, during operations, when the marine vessel **10** is not at berth (such as when marine vessel **10** is moving across or within a body of water), exhaust emitted from the outlets **54** of exhaust pipes **52** may flow into the enclosure **112** and then is emitted from the enclosure **112** via the one or more first flow paths **122**. Specifically, when the marine vessel **10** is not at berth, the valve **154** positioned along the collection pipe **150** may be transitioned to the closed position to prevent exhaust from flowing out of enclosure **112** and toward the deck **12** via the coupling **152**. As a result, the exhaust emitted from the outlets **54** of exhaust pipes **52** may flow out of the enclosure **112** to the surrounding atmosphere **123** via the vent(s) **120** and/or the vent(s) **160** along the first flow path(s) **122**. As previously described, the vents **120**, **160** may be actuated (e.g., via controller, personnel, etc.) to the open position to allow the exhaust to flow out of the enclosure **112** to the surrounding atmosphere **123**. Alternatively, for embodiments in which the vents **120**, **160** are pressure-actuated as previously described, the emission of the exhaust from the outlets **54** of exhaust pipes **52** into enclosure **112** increases the pressure within the enclosure **112** (e.g., such as when the valve **154** is closed) until one or more of the vent(s) **120**, **160** are transitioned to the open position to vent the exhaust to the surrounding atmosphere **123** along the first flow path(s) **122**.

As shown in FIGS. 4 and 7, when the marine vessel **10** is at berth (for example, is moored or docked at a berthing location as previously described), the exhaust emitted into the enclosure **112** via the outlets **54** of exhaust pipes **52** may be flowed out of the enclosure **112** via the second flow path **124**. Specifically, when the marine vessel **10** is at berth, an exhaust cleaning assembly **200** may be connected to the coupling **152** of the collection pipe **150** via a conduit **202**, and valve **154** (FIG. 4) may be transitioned to the open position. As a result, exhaust emitted from outlets **54** of exhaust pipes **52** may be flowed out of the enclosure **112** to the exhaust cleaning assembly **200** via the second flow path **124** and conduit **202**. As is known by one having ordinary skill in the art, within the exhaust cleaning assembly **200**, the exhaust may be treated via one or more suitable processes or

11

assemblies to remove some or all of the pollutants or other harmful constituents of the exhaust (e.g., nitrous oxide (NOx), carbon dioxide (CO₂), carbon monoxide (CO), soot, etc.). In some embodiments, the exhaust cleaning assembly 200 may simply capture and store the exhaust such that the exhaust may then be transferred to a suitable cleaning process or other storage facility. Thus, in some embodiments, the exhaust cleaning assembly 200 may include one or more tanks (or other suitable vessels 204 that are configured to receive and store the exhaust (or some other fluid such as treated exhaust, treatment fluids, etc.). Additionally or alternatively, in some embodiments, the exhaust cleaning system 200 may include one or more reactors 206 such as catalyst reactors, that are configured to treat or clean the exhaust during operations.

Regardless of the particular makeup or function of the exhaust cleaning assembly 200 (such whether the exhaust cleaning assembly 200 is configured to treat or simply store exhaust during operations), the exhaust emitted from the outlets 54 of exhaust pipes 52 may be prevented from flowing (or mostly restricted from flowing) to the atmosphere 123 at the berthing location when coupling 152 is connected to exhaust cleaning assembly 200 via conduit 202. For instance, when the exhaust is flowing along the second flow path 124 to the exhaust cleaning assembly, the vent(s) 120, 160 may be actuated to the closed position (such as via a controller or personnel or by a differential pressure between the atmosphere 123 and enclosure 112/collection pipe 150) to prevent a flow of exhaust along the first flow path(s) 122. As a result, the atmospheric conditions at the berthing location may be improved without having to shut down the exhaust-emitting systems 60 (FIG. 2) of the marine vessel 10.

As shown in FIG. 7, in some embodiments, the exhaust cleaning assembly 200 may be positioned on at barge or other marine vessel that is positioned proximate to (e.g., alongside) the marine vessel 10 while the marine vessel 10 is at the berthing location. As shown in FIG. 8, in some embodiments, the exhaust cleaning assembly 200 may be positioned onshore at the berthing location. In some embodiments, the exhaust cleaning assembly 200 (or a portion thereof) may be remote from the berthing location (e.g., one or more miles away from the berthing location) and the conduit 202 may include (or be connected to) a suitable pipeline or other suitable infrastructure to transport the exhaust from the berthing location to the remotely located exhaust cleaning assembly 200.

As shown in FIG. 9, as previously described, in some embodiments, exhaust is output from the outlets 54 of the exhaust pipes 52 into the enclosure 112, and then the exhaust may be flowed out of the enclosure 112 via the one or more first flow paths 122 or one or more second flow paths 124. However in some embodiments, the collection pipe 150 may be individually connected to the outlets 54 of one or more of the exhaust pipes 52 via a plurality of pipe connections 158. The pipe connections 158 (or more simply "connections" 158) may include any suitable conduit and/or connector that may be connected to and thus in fluid communication with the outlets 54 to the collection pipe 150. For instance, in some embodiments, the pipe connections 158 may include pipes, hoses, tubing, or some combination thereof. During operations with the embodiment depicted in FIG. 9, exhaust emitted from the outlets 54 of the exhaust pipes 52 may flow directly into the collection pipe 150 via the plurality of pipe connections 158 and may thus bypass (and not flow into) the enclosure 112 defined by cap 110. In some embodiments, the pipe connection 158 may route exhaust from one or more of

12

the exhaust pipes 52 separately from exhaust from one or more others of the exhaust pipes 52. For instance, one or more of the exhaust pipes 52 may output exhaust to an exhaust cleaning system 200 via the pipe connections 158 and collection pipe 150 and one or more others of the exhaust pipes 52 may output exhaust to the atmosphere 123 and/or another exhaust cleaning system 200 via the enclosure 112 and another collection pipe 150 (not shown) connected to cap 110.

As is shown in FIG. 9, for embodiments that include pipe connections 158 for individually connecting the outlet 54 of exhaust pipes 52 to collection pipe 150, the one or more first flow paths 122 may extend from the collection pipe 150 through one or more vents 120 (or one or more vents 160) connected to collection pipe 150. Thus, in these embodiments, the first flow path(s) 122 may bypass the enclosure 112. Vent(s) 120, 160 may also be positioned on the cap 110 (e.g., along planar top 116) to allow exhaust (or other fluid) to vent from enclosure 112 (e.g., in the event the exhaust is leaking from pipes 52, connections 158, collection pipe 150, etc.). In addition, for the embodiment of FIG. 9, the second flow path 124 may extend through the collection pipe 150 toward the deck 12 and coupling 152 as shown in FIG. 4 and previously described.

As shown in FIG. 10, in some embodiments, each pipe connection 158 may independently route the exhaust emitted from the corresponding exhaust pipe 52 out of the enclosure 112 of cap 110 via a separate collection pipe 159. Thus, in these embodiments, the common collection pipe 150 may be omitted. However, in some embodiments, each of the separate collection pipes 159 (or some of the collection pipes 159) may be nested within the common collection pipe 150. As shown in FIG. 10, in some embodiments, the separate collection pipes 159 may independently route exhaust from the outlets 54 of exhaust pipes 52 to the surrounding atmosphere 123 or to one or more exhaust cleaning assemblies (such as, exhaust cleaning assembly 200). For instance, within each of the separate collection pipes 159, a first flow path 122 may be defined that routes exhaust from the corresponding collection pipe 159 and out to the surrounding atmosphere 123 via a vent 120 that is connected to the corresponding collection pipe 159. In addition, within each of the separate collection pipes 159, a second flow path 124 may be defined that routes exhaust from the corresponding collection pipe 159 toward a coupling 152 connected to the corresponding collection pipe 159 and accessible from the deck 12 of the marine vessel 10 (FIG. 4). Each of the couplings 152 (or some of the couplings 152) connected to the separate collection pipes 159 may be connected (such as, via a conduit such as conduit 202 shown in FIGS. 4 and 7) to an exhaust cleaning assembly (e.g., exhaust cleaning assembly 200 shown in FIGS. 4 and 5) when the marine vessel 10 is at a berthing location as previously described.

Reference is now generally made to FIGS. 11A-11D, in which an example sequence or method for routing exhaust from the marine vessel 10 via the exhaust handling system 100 when the marine vessel 10 is positioned at a berthing location 250 (such that the marine vessel 10 is berthed) and when the marine vessel 10 is traveling into and out of the berthing location 250 is shown according to some embodiments. In describing the sequence shown in FIGS. 11A-11D, continuing reference will be made to the various features of embodiments of marine vessel 10 and exhaust handling system 100 shown in FIGS. 1-10 and previously described above.

Initially, as shown in FIGS. 11A and 11B, the marine vessel 10 may travel toward the berthing location 250. For

13

instance, as previously described, the berthing location 250 may include a port, dock, harbor, mooring, etc., and the marine vessel 10 may travel to the berthing location 250 for any suitable reason (such as to offload or receive cargo, undergo repairs, refuel, offload or receive personnel, undergo inspection, etc.). When the marine vessel 10 is traveling (such as across a body of water such an ocean, lake, channel, etc.), the exhaust emitted from exhaust stack 50 (particularly from the exhaust pipes 52 extending through exhaust stack 50), may be vented to the surrounding atmosphere 123 via the first flow path(s) 122 (FIG. 4) extending out of the cap 110. An exhaust cleaning assembly 200 may be positioned at (or accessible from) the berthing location 250 as previously described. For instance, as previously described, the exhaust cleaning assembly 200 may be positioned on a barge that is floating on the water within the berthing location 250, or the exhaust cleaning assembly 200 may be positioned onshore at the berthing location 250 (or at a remote location from the berthing location 250 and accessible via pipeline or other suitable infrastructure as previously described).

As shown in FIG. 11C, once the marine vessel 10 is positioned within the berthing location 250, the exhaust handling system 100 may be fluidly connected to the exhaust cleaning assembly 200 via coupling 152 and conduit 202 as previously described. As a result, while the marine vessel 10 is berthed, any exhaust emitted from the exhaust stack 50 (particularly from the one or more exhaust pipes 52) may be flowed to the exhaust cleaning assembly 200 via the exhaust handling system 100 to be stored (such as in the one or more tanks 204), cleaned (such as via the one or more reactors 206), or otherwise processed as previously described. Thereafter, as shown in FIG. 11D, when the time comes for marine vessel 10 to depart from the berthing location 250, the exhaust cleaning assembly 200 may be disconnected from the exhaust handling system 100, and the marine vessel 10 may travel out of and away from the berthing location 250. Once the exhaust handling system 100 is disconnected from exhaust cleaning assembly 200 and while marine vessel 10 is traveling out of and away from the berthing location 250, any exhaust emitted from exhaust stack 50 (particularly the one or more exhaust pipes 52) may once again be emitted to the surrounding atmosphere 123 via the first flow path(s) 122 (FIG. 4) extending out of the cap 110.

Thus, exhaust emitted from the exhaust stack 50 of marine vessel 10 may be continuously vented via the exhaust handling system 100 both when the marine vessel 10 is berthed (such as at the berthing location 250) and when the marine vessel 10 is away from the berthing location 250 (and traveling to and away therefrom). By permanently fixing and integrating the exhaust handling system 100 (including cap 110 and collection pipe 150) onto the marine vessel 10, personnel may simply connect the conduit 202 to the coupling 152 upon arrival at the berthing location 250, thereby eliminating the need to use heavy-lift equipment, such as a crane, to transfer or place a temporary cap atop the exhaust stack 50 once the marine vessel 10 arrives at the berthing location 250. Accordingly, when the marine vessel 10 is berthed (such as at a berthing location) exhaust-emitting systems (such as exhaust-emitting systems 60 shown in FIG. 2) may continue to operate so as to support operation of the marine vessel 10 and any sub-systems thereof, and exhaust may be prevented (or restricted) from being emitted into the surrounding atmosphere 123 within the berthing location 250. As a result, the air quality of a berthing location 250 may be maintained, even is multiple marine vessels 10 are positioned therein.

14

As shown in FIG. 12, in some embodiments, a controller 300 may be used to control the flow of exhaust through the exhaust handling system 100 during operation of the marine vessel 10 both when the marine vessel 10 is berthed (for example, at berthing location 250 shown in FIGS. 11A-11D) and when the marine vessel 10 is not berthed. The controller 300 may be (or may be incorporated within) a main or master controller onboard the marine vessel 10, or the controller 300 may be a standalone controller 300 for controlling the flow of exhaust through the exhaust handling system 100. In either case, the controller 300 may be described and referred to herein as being a part of the exhaust handling system 100. The controller 300 may be positioned onboard the marine vessel 10, such as on the bridge 20; however, controller 300 may be positioned at any location (or distributed among multiple locations) onboard or off the marine vessel 10.

The controller 300 may be a computing device, such as a computer, tablet, smartphone, server, or other computing device or system. Thus, controller 300 may include a processor 302 and a memory 304. The processor 302 may include any suitable processing device or a collection of processing devices. In some embodiments, the processor 302 may include a microcontroller, central processing unit (CPU), graphics processing unit (GPU), timing controller (TCN), scaler unit, or some combination thereof. During operations, the processor 302 executes machine-readable instructions (such as machine-readable instructions 306) stored on memory 304, thereby causing the processor 302 to perform some or all of the actions attributed herein to the controller 300. In general, processor 302 fetches, decodes, and executes instructions (e.g., machine-readable instructions 306). In addition, processor 302 may also perform other actions, such as, making determinations, detecting conditions or values, etc., and communicating signals. If processor 302 assists another component in performing a function, then processor 302 may be said to cause the component to perform the function.

The memory 304 may be any suitable device or collection of devices for storing digital information including data and machine-readable instructions (such as machine-readable instructions 306). For instance, the memory 304 may include volatile storage (such as random access memory (RAM)), non-volatile storage (e.g., flash storage, read-only memory (ROM), etc.), or combinations of both volatile and non-volatile storage. Data read or written by the processor 302 when executing machine-readable instructions 306 can also be stored on memory 304. Memory 304 may include "non-transitory machine-readable medium," where the term "non-transitory" does not include or encompass transitory propagating signals.

The processor 302 may include one processing device or a plurality of processing devices that are distributed within controller 300 or more broadly within marine vessel 10. Likewise, the memory 304 may include one memory device or a plurality of memory devices that are distributed within controller 300 or more broadly within marine vessel 10.

The controller 300 may be communicatively connected (such as via wired and/or wireless connection) to a user interface 308 (such as a monitor, display, computing device, touch-sensitive screen or other surface, keyboard, mouse, or some combination thereof). During operations, a user (e.g., personnel onboard the marine vessel 10) may view information output from the controller 300 on the user interface 308 (such as the position or status of one or more of the sensors 320, 322, vent(s) 120, vent(s) 160, valve 154, coupling 152, etc.). In addition, during operations, a user

15

may make inputs to the controller 300 via the user interface 308 (such as commands to open valve 154 and/or vent(s) 120, 160).

Controller 300 may be connected to various sensors (such as sensors 320, 322) positioned throughout the exhaust handling system 100. For instance, controller 300 may be connected to a pressure sensor 320 that is connected to the cap 110 and configured to detect or measure a pressure within the enclosure 112 (or value indicative thereof). In addition, controller 300 may be connected to a pressure sensor 322 that is connected to the collection pipe 150 and configured to detect or measure a pressure within the collection pipe 150 (or a value indicative thereof). The pressure sensors 320, 322 may be any suitable device that is configured to measure, detect, or determine a pressure (or value indicative thereof) within a given area, volume, location. For instance, in some embodiments, the pressure sensors 320, 322 may include strain gauges, capacitance-based pressure sensors, solid-state pressure sensors, manometers, barometers, resistive pressure sensors, etc.

In addition, controller 300 may be connected to the valve 154 positioned along collection pipe 150 and the one or more vents 120, 160 (note: one vent 120 positioned along collection pipe 150 and one vent 160 positioned on the cap 110 are shown in FIG. 12, but any one or more of the vents 120 and/or vents 160 shown in FIG. 4 may be included and connected to controller 300 in some embodiments). For instance, the valve 154 and/or the one or more vents 120, 160 may include motorized or actuatable valves that may be transitioned between the open and closed positions (previously described above) via the controller 300 (or a command or signal generated thereby). In addition, controller 300 may be connected to the valve 154 and/or the one or more vents 120, 160 such that controller 300 may determine a position (such as open, closed, or a position between fully open or fully closed) of the valve 154 and/or the one or more vents 120, 160. For instance, the valve 154 and/or the one or more vents 120, 160 may include or be connected to a driver such as a stepper motor, servo-motor, or other suitable device that may controllably place the valve 154 and/or the one or more vents 120, 160 in a particular position. The controller 300 may determine a position of the valve 154 and/or the one or more vents 120, 160 via the previous actuation(s) of the drivers (e.g., stepper motor, servo-motor, etc.). In some embodiments, one or more additional sensors or devices, such as position sensors, proximity sensors, pressure sensors (e.g., differential pressure sensors), optical sensors, etc. may be utilized by controller 300 to determine a position of the valve 154 and/or the one or more vents 120, 160 during operations.

Further, controller 300 may be connected to a sensor 324 that is configured to detect whether a conduit (such as conduit 202 shown in FIGS. 7, 8, and 11C) is connected to the coupling 152 of collection pipe 150. The sensor 324 may include a proximity sensor, optical sensor, magnetic sensor, switch, or any other suitable device that may be configured to detect when a conduit (or coupling connected thereto) is engaged with the coupling 152 such that exhaust flowing out of the collection pipe 150 may be directed to another location or system (such as exhaust cleaning assembly 200 shown in FIGS. 7, 8, and 11A-11C).

During operations, controller 300 may selectively actuate the valve 154 and/or the vent(s) 120, 160 between the open and closed positions so as to route the exhaust emitted from the one or more exhaust pipes 52 either via the first flow path(s) 122 or the second flow path 124. For instance, when marine vessel 10 is at berth (e.g., such as shown in FIGS. 9B

16

and 9C previously described above), and a conduit (e.g., conduit 202) is engaged with the coupling 152, the controller 300 may open the valve 154 and close the vent(s) 120, 160 so that exhaust emitted from the one or more exhaust pipes 52 may be directed along the second flow path 124 and out of the coupling 152 and into an exhaust cleaning assembly (such as exhaust cleaning assembly 200) as previously described. In addition, when an exhaust cleaning assembly is not connected to the collection pipe 150 (such as via a conduit connected to coupling 152), such as when the marine vessel 10 is not at berth, the controller 300 may close the valve 154 and open one or more of the vent(s) 120, 160 to allow exhaust emitted from the one or more exhaust pipes 52 to be vented to the surrounding atmosphere 123. Further, in some embodiments, regardless as to whether the marine vessel 10 is at berth or not at berth (and thus regardless as to whether an exhaust cleaning assembly is connected to the collection pipe 150 via coupling 152) the controller 300 may actuate one or more of the vents 120, 160 to open (and thereby vent exhaust to the surrounding atmosphere 123) in response to a determination (e.g., by the controller 300) that the pressure within the enclosure 112 and/or within the collection pipe 150 is above a threshold. The controller 300 may determine the pressure within the enclosure 112 and within the collection pipe 150 based on outputs received from the sensors 320, 322. In some embodiments, one or more temperature sensors (e.g., a thermocouple, thermistor, resistance temperature detector (RTD), semiconductor circuit, etc.) may be connected to the exhaust handling system 100 (such as to enclosure 112, collection pipe 150, etc.), and the controller 300 (alternatively or additionally) may actuate the one or more vents 120 to open (and thereby vent exhaust to the atmosphere 123) in response to a determination (such as by the controller 300) that the temperature within the enclosure 112 and/or the collection pipe 150 is above a threshold.

As shown in FIGS. 10 and 12, in some embodiments, the exhaust pipes 52 (or some of the exhaust pipes 52) may be connected to separate collection pipes 159 that are further connected to separate vents 120, 160, valves 154, and couplings 152 as previously described. In these embodiments, the controller 300 may be connected to one or more pressure and/or temperature sensors (such as pressure sensors 320, 322) that are configured to detect a pressure and/or temperature within the separate collection pipes 159. In addition, the controller 300 may be connected to the couplings 152 and valves 154 of the separate collection pipes 159. During operations, the controller 300 may direct exhaust through either the first flow path 122 or the second flow path 124 within each of the separate exhaust pipes 159 via actuation of the separate valves 154 and vent(s) 120, 160 in a similar manner to that previously described for the common collection pipe 150 shown in FIG. 12.

In some embodiments the vent(s) 120, 160 may be pressure-actuated as previously described. Thus, during operations, one or more of the vents 120, 160 may open or close based on a pressure within the enclosure 112 or collection pipe 150 (or the separate collection pipes 159) regardless of a position of the valve 154 and/or connection status of the coupling 152 (such as whether the coupling 152 is connected to or not connected to a conduit of an exhaust cleaning assembly such as the conduit 202 and exhaust cleaning assembly 200 shown in FIGS. 7 and 8). Thus, in some embodiments, the controller 300 may not be connected to the one or more vent(s) 120, 160 of the exhaust handling system 100.

Reference is now made to FIGS. 13 and 14, in which methods 400, 450 for flowing the exhaust of a marine vessel through an exhaust handling system (such as the exhaust handling system 100) are shown according to some embodiments. The method 400 may include a method for flowing the exhaust emitted from the marine vessel (such as marine vessel 10) along a first flow path to the atmosphere or a second flow path to an exhaust cleaning assembly (such as exhaust cleaning assembly 200 shown in FIGS. 7 and 8), and the method 450 may include a method for flowing exhaust to the atmosphere based on a pressure within the exhaust handling system (such as exhaust handling system 100). In some embodiments, the methods 400, 450 may be performed (wholly or partially) by the processor of a controller (such as processor 302 of controller 300 shown in FIG. 12). Thus, the methods 400, 450 may be representative of the machine-readable instructions 306 stored on memory 304, or some of the machine-readable instructions 306 for some embodiments of controller 300 (FIG. 12). In addition, in describing the features of methods 400, 450, continuing reference will be made to the features of the embodiments shown in FIGS. 1-12 and previously described. In some embodiments, the methods 400, 450 may be performed in parallel (such as by processor 302), and/or may be combined or integrated with one another.

The method 400 shown in FIG. 13 may initially include a determination, at block 402, as to whether a coupling of a collection pipe of an exhaust handling system (such as exhaust handling system 100) is connected to an exhaust cleaning assembly (such as exhaust cleaning assembly 200). For instance, as shown in FIG. 12 and previously described, the controller 300 may determine whether a conduit 202 of an exhaust cleaning assembly 200 (FIGS. 7 and 8) is connected to the coupling 152 of collection pipe 150 via an output from the sensor 324 as previously described.

If it is determined that the coupling of the collection pipe is not connected to an exhaust cleaning assembly (the determination at block 402 is “No” or “N”), the method 400 may proceed to flow exhaust emitted from the marine vessel 10 along a first flow path to the atmosphere (for example, atmosphere 123) at block 404. For instance, as shown in FIG. 12 and previously described, in some embodiments, the controller 300 may determine that a conduit 202 of an exhaust cleaning assembly 200 is not connected to coupling 152 of collection pipe 150 via an output from the sensor 324, and may thus close the valve 154 and open one or more of the vent(s) 120, 160 (or allow the vent(s) 120, 160 to open via pressure actuation as previously described) so as to flow the exhaust emitted from the one or more exhaust pipes 52 into the surrounding atmosphere 123 via the one or more first flow paths 122.

Conversely, if it is determined that the coupling of the collection pipe is connected to an exhaust cleaning assembly (the determination at block 402 is “Yes” or “Y”), the method 400 may proceed to flow exhaust emitted from the marine vessel 10 along a second flow path to the exhaust cleaning assembly at block 406. For instance, as shown in FIG. 12 and previously described, in some embodiments, the controller 300 may determine that a conduit 202 is connected is connected to the coupling 152 via an output from the sensor 324, and may thus may open the valve 154 and (potentially) close the vent(s) 120, 160 so as to flow the exhaust emitted from the one or more exhaust pipes 52 along the second flow path 124 to the exhaust cleaning assembly 200.

The method 450 shown in FIG. 14 may initially include receiving an indication of a pressure within an enclosure and/or a collection pipe of an exhaust handling system (such

as exhaust handling system 100) at block 452. For instance, as shown in FIG. 12 and previously described, the controller 300 may receive an indication of a pressure within the enclosure 112 and/or the collection pipe 150 via outputs from the pressure sensor 320 and/or the pressure sensor 322, respectively.

Next, method 450 includes determining whether the pressure (such as the pressure within the enclosure 112 and/or the pressure within the collection pipe 150) is above a threshold at block 454. For instance, block 454 may include determining whether a particular one or either of the pressures within the enclosure (e.g., enclosure 112) of the collection pipe (e.g., collection pipe 150) is above a threshold. In some embodiments, block 454 may include determining whether the pressure within the enclosure (for example, enclosure 112) is above a first threshold and/or whether the pressure within the collection pipe (for example, collection pipe 150) is above a second threshold, wherein the first and second thresholds are different. The threshold (or thresholds) may be determined based upon an operating pressure (or pressure range) of the enclosure (for example, enclosure 112) and/or the collection pipe (e.g., collection pipe 150).

If it is determined that a pressure within the enclosure and/or the collection pipe is above a threshold (the determination at block 454 is “Yes” or “Y”), method 450 may proceed to open one or more vents to vent exhaust from the enclosure and/or the collection pipe block 456. For instance, as shown in FIG. 12 and previously described, if the controller 300 determines that the pressure within the exhaust handling system 100 (including the enclosure 112 and/or the collection pipe 150) is above a threshold, the controller 300 may open one or more of the vent(s) 120 or the vent(s) 160 so as to flow the exhaust out of the enclosure 112 and/or collection pipe 150 to the surrounding atmosphere 123 in an effort to reduce the pressure below the threshold. Conversely, if it is determined that a pressure within the enclosure and/or the collection pipe is not above a threshold (the determination at block 454 is “No” or “N”), method 450 may repeat block 452 and continue monitoring the pressure within the enclosure and/or the collection pipe.

If the vent(s) (for example, vent(s) 120, vent(s) 160) are open via block 456, method 450 may proceed to once again receive an indication of the pressure within the enclosure and/or the collection pipe at block 457 and then determine whether the pressure within the enclosure and/or the collection pipe 150 is below a threshold at block 458. The indication of the pressure received at block 457 may be obtained in a similar manner to that described above for block 452. The threshold at block 458 may be same or different from the threshold in block 454. In some embodiments, the threshold at block 458 may be lower than the threshold at block 454. The threshold at block 458 may be determined such that the pressure within the enclosure and/or the collection pipe is within a desired or operating range (potentially including a safety factor). If it is determined that the pressure is below the threshold at block 458 (the determination at block 458 is “Yes” or “Y”), method 450 may proceed to close the vent(s) at block 460. For instance, as shown in FIG. 12 and previously described, if the controller 300 determines that the pressure within the enclosure 112 and/or collection pipe 150 is below a threshold (or within a desired range) (such as via the sensors 320, 322) after the vent(s) 120 and/or the vent(s) 160 have been opened to direct exhaust to the atmosphere 123, the controller 300 may close the vent(s) 120, 160. In some embodiments, closing the vent(s) 120, 160 may allow the exhaust to

19

flow along the second flow path **124** toward the exhaust cleaning assembly (such as exhaust cleaning assembly **200**) as previously described.

Conversely, if it is determined that the pressure within the enclosure and/or the collection pipe is not below the threshold at block **458** (the determination at block **458** is “No” or “N”), method **450** may repeat back to block **457** to once again receive an indication of the pressure, and then determine whether the pressure within the enclosure and/or the collection pipe is below the threshold at block **458**. For instance, as shown in FIG. **12** and previously described, if after opening the vent(s) **120**, **160**, the controller **300** determines is still above a threshold (and is therefore above a desired range), the controller **300** may maintain the vent(s) **120**, **160** in the open position and continue to monitor the pressure within the enclosure **112** and/or the collection pipe **150** via the sensors **320**, **322** as previously described.

As explained above and reiterated below, this disclosure includes, without limitation, the following example embodiments.

Example Embodiment 1: an exhaust handling system for a marine vessel, the exhaust handling system comprising: a cap connected to a top end portion of an exhaust stack of the marine vessel so as to form an enclosure that at least partially surrounds an outlet of an exhaust pipe extending through the exhaust stack; a collection pipe in fluid communication with the cap such that the collection pipe is configured to receive exhaust from the enclosure; and a coupling connected to the collection pipe that is configured to connect to an exhaust cleaning assembly, the exhaust cleaning system including at least one tank to receive the exhaust, the cap at least partially defines a first flow path for the exhaust to flow from the enclosure to an atmosphere surrounding the cap, and the collection pipe at least partially defines a second flow path for the exhaust to flow from the enclosure to the coupling via the collection pipe.

Example Embodiment 2: the exhaust handling system of any example embodiment, wherein the cap is connected to the exhaust stack such that the cap is supported by the exhaust stack.

Example Embodiment 3: the exhaust handling system of any example embodiment, wherein the cap is welded to the exhaust stack.

Example Embodiment 4: the exhaust handling system of any example embodiment, further comprising a vent, connected to the cap, wherein the vent at least partially defines the first flow path, and wherein the vent includes at least one valve member that is configured to actuate between: an open position to emit exhaust to the atmosphere via the first flow path; and a closed position to prevent emission of exhaust to the atmosphere via the first flow path.

Example Embodiment 5: the exhaust handling system of any example embodiment, wherein the vent comprises a pressure-actuated vent.

Example Embodiment 6: the exhaust handling system of any example embodiment, wherein the at least one valve member comprises a plurality of louvers that are rotatable to transition the vent between the open position and the closed position.

Example Embodiment 7: the exhaust handling system of any example embodiment, wherein the coupling is proximate to a deck of the marine vessel.

Example Embodiment 8: the exhaust handling system of any example embodiment, further comprising a cleanout port positioned along the collection pipe that is

20

proximate to the deck of the marine vessel, wherein the cleanout port is configured to provide access into the collection pipe independent of the coupling.

Example Embodiment 9: the exhaust handling system of any example embodiment, further comprising a pipe connection positioned within the enclosure that is in fluid communication with the exhaust pipe such that exhaust pipe is configured to output exhaust into the pipe connection.

Example Embodiment 10: the exhaust handling system of any example embodiment, wherein the pipe connection is connected to the collection pipe such that exhaust emitted from the exhaust pipe is routed into the collection pipe via the pipe connection.

Example Embodiment 11: the exhaust handling system of any example embodiment, wherein the collection pipe comprises flexible ductwork.

Example Embodiment 12: an exhaust handling system for a marine vessel, the exhaust handling system comprising: a cap connected to a top end portion of an exhaust stack of the marine vessel so as to form an enclosure that at least partially surrounds an outlet of an exhaust pipe extending through the exhaust stack; a vent in fluid communication with the enclosure, the vent including at least one valve member that is actuatable between a first position to emit exhaust from the enclosure to an atmosphere surrounding the cap via the vent and a second position to prevent an emission of exhaust from the enclosure to the atmosphere via the vent; and a collection pipe in fluid communication with the enclosure; a coupling connected to the collection pipe and configured to connect to an exhaust cleaning assembly that includes at least one tank to receive the exhaust.

Example Embodiment 13: the exhaust handling system of any example embodiment, wherein the at least one valve member comprises a plurality of louvers that are rotatable to transition the vent between the first position and the second position.

Example Embodiment 14: the exhaust handling system of any example embodiment, wherein the vent comprises a pressure-actuated vent.

Example Embodiment 15: the exhaust handling system of any example embodiment, wherein the pressure-actuated vent is positioned on a top end of the cap.

Example Embodiment 16: the exhaust handling system of any example embodiment, wherein the pressure-actuated vent is positioned along the collection pipe.

Example Embodiment 17: the exhaust handling system of any example embodiment, wherein the cap is connected to the exhaust stack such that the cap is supported by the exhaust stack.

Example Embodiment 18: the exhaust handling system of any example embodiment, wherein the cap is welded to the exhaust stack.

Example Embodiment 19: the exhaust handling system of any example embodiment, wherein the coupling is proximate to a deck of the marine vessel.

Example Embodiment 20: the exhaust handling system of any example embodiment, wherein the collection pipe comprises a cleanout port that is proximate to the deck of the marine vessel, and wherein the cleanout port is configured to provide access into the collection pipe independent of the coupling.

Example Embodiment 21: the exhaust handling system of any example embodiment, further comprising a pipe connection positioned within the enclosure that is in fluid communication with the exhaust pipe and the

21

collection pipe such that the exhaust pipe is configured to output exhaust into collection pipe via the pipe connection.

Example Embodiment 22: an exhaust handling system for a marine vessel, the exhaust handling system comprising: a cap connected to a top end portion of an exhaust stack of the marine vessel such that the cap is supported by the exhaust stack and such that the cap forms an enclosure that at least partially surrounds an outlet of an exhaust pipes extending through the exhaust stack; a pressure-actuated vent in fluid communication with the enclosure; and a collection pipe in fluid communication with the enclosure and configured to connect to an exhaust cleaning assembly that includes at least one tank to receive the exhaust.

Example Embodiment 23: the exhaust handling system of any example embodiment, wherein the pressure-actuated vent includes at least one valve member that is actuable to selectively emit exhaust from the enclosure to an atmosphere surrounding the cap.

Example Embodiment 24: the exhaust handling system of any example embodiment, wherein the pressure-actuated vent is positioned on a top end of the cap.

Example Embodiment 25: the exhaust handling system of any example embodiment, wherein the pressure-actuated vent is positioned along the collection pipe.

Example Embodiment 26: the exhaust handling system of any example embodiment, wherein the cap is welded to the exhaust stack.

Example Embodiment 27: the exhaust handling system of any example embodiment, further comprising a coupling connected to the collection pipe that is configured to connect to a conduit of the exhaust cleaning assembly, wherein the coupling is accessible from a deck of the marine vessel.

Example Embodiment 28: the exhaust handling system of any example embodiment, wherein the collection pipe comprises a cleanout port that is accessible from the deck of the marine vessel, and wherein the cleanout port is configured to provide access into the collection pipe independent of the coupling.

Example Embodiment 29: the exhaust handling system of any example embodiment, further comprising a pipe connection positioned within the enclosure that is in fluid communication with the exhaust pipe such that the exhaust pipe is configured to output exhaust to the collection pipe via the pipe connection.

Example Embodiment 30: a method comprising: (a) positioning a marine vessel in a berthing location, the marine vessel including a deck, an exhaust stack having an top end portion positioned above from the deck, an exhaust pipe extending through the exhaust stack to an outlet that is positioned above the top end portion of the exhaust stack, a cap connected to the exhaust stack so as to form an enclosure that at least partially surrounds the outlet of the exhaust pipe, and an collection pipe in fluid communication with the enclosure; (b) connecting an exhaust cleaning assembly to the collection pipe after (a), the exhaust cleaning assembly including at least one tank to receive the exhaust; and (c) operating the marine vessel to flow an exhaust emitted from the outlet of the exhaust pipe to the exhaust cleaning assembly after (b) via the collection pipe.

Example Embodiment 31: the method of any example embodiment, further comprising: (d) emitting the exhaust out of the enclosure to an atmosphere surrounding the cap during (a).

22

Example Embodiment 32: the method of any example embodiment, wherein (d) further comprises flowing the exhaust out of a vent in fluid communication with the enclosure.

Example Embodiment 33: the method of any example embodiment, wherein (d) further comprises flowing the exhaust out of a pressure-actuated vent in fluid communication with the enclosure.

Example Embodiment 34: the method of any example embodiment, further comprising: (e) disconnecting the exhaust cleaning assembly from the collection pipe; (f) moving the marine vessel away from the berthing location after (e); and (g) flowing the exhaust out of the enclosure to the atmosphere during (f).

Example Embodiment 35: the method of any example embodiment, wherein (b) comprises connecting a conduit connected to the exhaust cleaning assembly to a coupling that is connected to the collection pipe and that is proximate to the deck of the marine vessel.

Example Embodiment 36: the method of any example embodiment, further comprising preventing the exhaust from flowing from the enclosure to an atmosphere surrounding the cap during (c).

Example Embodiment 37: the method of any example embodiment, wherein the exhaust cleaning assembly is positioned on a barge that is further positioned at the berthing location.

Example Embodiment 38: a method comprising: (a) positioning a marine vessel in a berthing location, the marine vessel including an exhaust stack, an exhaust pipe extending through the exhaust stack to an outlet that is positioned above a top end portion of the exhaust stack, a cap that is connected to the exhaust stack to form an enclosure that surrounds the outlet of the exhaust pipe; (b) flowing an exhaust from the enclosure to an atmosphere surrounding the cap via a first flow path during (a), the first flow path at least partially defined by the cap; (c) connecting an exhaust cleaning assembly positioned at the berthing location to the cap after (a), the exhaust cleaning system including at least one tank to receive the exhaust; and (d) flowing the exhaust from the enclosure to the exhaust cleaning assembly via a second flow path after (b), the second flow path at least partially defined by the cap.

Example Embodiment 39: the method of any example embodiment, wherein (b) further comprises flowing the exhaust out of a vent connected to the enclosure.

Example Embodiment 40: the method of any example embodiment, wherein (b) further comprises rotating at least one louver of the vent to open the vent.

Example Embodiment 41: the method of any example embodiment, further comprising: (e) disconnecting the exhaust cleaning assembly from the cap; (f) moving the marine vessel away from the berthing location after (e); and (g) flowing the exhaust out of the enclosure to the atmosphere via the first flow path during (f).

Example Embodiment 42: the method of any example embodiment, wherein (c) comprises connecting a conduit connected to the exhaust cleaning assembly to a coupling that is proximate to a deck of the marine vessel and that is in fluid communication with the enclosure.

Example Embodiment 43: the method of any example embodiment, further comprising preventing the exhaust from flowing from the enclosure to the atmosphere during (d).

Example Embodiment 44: the method of any example embodiment, wherein the exhaust cleaning assembly is positioned on a barge that is positioned at the berthing location.

As described above, the embodiments disclosed herein include exhaust handling systems for a marine vessel that are configured to allow selective collection of exhaust output from the exhaust-emitting systems of the marine vessel while at a berthing location. In some embodiments, the collected exhaust may be routed to an exhaust cleaning assembly that may store and/or process the exhaust to prevent harmful chemicals or pollutants therein from being emitted to the atmosphere. In some embodiments, the exhaust handling systems of the embodiments disclosed herein may include a cap that is connected to an upper end portion of an exhaust stack of the marine vessel so as to form an enclosure around one or more (e.g., a plurality of) exhaust pipes that emit exhaust from the exhaust-emitting systems of the marine vessel. The collected exhaust may be emitted directly to the atmosphere when the marine vessel is not at a berthing location (or during a situation that requires venting of exhaust to the atmosphere), or may be selectively routed to an exhaust cleaning assembly to avoid such atmospheric venting when the marine vessel is berthed. Accordingly, through use of the embodiments disclosed herein, the exhaust-emitting systems of a marine vessel may continue to operate when the marine vessel is berthed while preventing (or restricting) the emission of exhaust (or at least the harmful and/or polluting components thereof) to the atmosphere.

The discussion above is directed to various exemplary embodiments. However, one of ordinary skill in the art will understand that the examples disclosed herein have broad application, and that the discussion of any embodiment is meant only to be exemplary of that embodiment, and not intended to suggest that the scope of the disclosure, including the claims, is limited to that embodiment.

The drawing figures are not necessarily to scale. Certain features and components herein may be shown exaggerated in scale or in somewhat schematic form and some details of conventional elements may not be shown in interest of clarity and conciseness.

In the discussion herein and in the claims, the terms “including” and “comprising” are used in an open-ended fashion, and thus should be interpreted to mean “including, but not limited to” Also, the terms “couple,” “couples,” “connect,” or “connection,” are intended to mean either an indirect or direct connection. Thus, if a first device couples or connects to a second device, that connection may be through a direct connection of the two devices, or through an indirect connection that is established via other devices, components, nodes, and connections. In addition, when used herein (including in the claims), the words “about,” “generally,” “substantially,” “approximately,” and the like, when used in reference to a stated value mean within a range of plus or minus 10% of the stated value. Further, as used herein, the terms “axial” and “axially” generally mean along or parallel to a given axis (e.g., central axis of a body or a port), while the terms “radial” and “radially” generally mean perpendicular to the given axis. For instance, an axial distance refers to a distance measured along or parallel to the axis, and a radial distance means a distance measured perpendicular to the axis.

This application is a continuation of U.S. Non-Provisional application Ser. No. 18/369,485, filed Sep. 18, 2023, titled “Exhaust Handling Systems for Marine Vessels and Related Methods,” which claims priority to and the benefit of U.S.

Provisional Application No. 63/485,886, filed Feb. 18, 2023, titled “Exhaust Handling Systems for Marine Vessels and Related Methods,” and U.S. Provisional Application Ser. No. 63/488,574, filed Mar. 6, 2023, titled “Exhaust Handling Systems for Marine Vessels and Related Methods,” the contents of which are incorporated herein by reference in their entirety.

While exemplary embodiments have been shown and described, modifications thereof can be made by one skilled in the art without departing from the scope or teachings herein. The embodiments described herein are exemplary only and are not limiting. Many variations and modifications of the systems, apparatus, and processes described herein are possible and are within the scope of the disclosure. Accordingly, the scope of protection is not limited to the embodiments described herein, but is only limited by the claims that follow, the scope of which shall include all equivalents of the subject matter of the claims. Unless expressly stated otherwise, the steps in a method claim may be performed in any order. The recitation of identifiers such as (a), (b), (c) or (1), (2), (3) before steps in a method claim are not intended to and do not specify a particular order to the steps, but rather are used to simplify subsequent reference to such steps.

What is claimed is:

1. An exhaust handling system for a marine vessel, the exhaust handling system comprising:

a cap connected to a top end portion of an exhaust stack of the marine vessel so as to define an enclosure positioned to at least partially surround an outlet of an exhaust pipe extending through the exhaust stack of the marine vessel;

a collection pipe in fluid communication with the cap and positioned to receive exhaust from the enclosure so that the cap at least partially defines a first flow path for the exhaust to flow from the enclosure to an atmosphere surrounding the cap;

a coupling connected to the collection pipe so that the collection pipe at least partially defines a second flow path for the exhaust to flow from the enclosure to the coupling via the collection pipe; and

an exhaust cleaning assembly connected to the coupling and including at least one tank to receive the exhaust.

2. The exhaust handling system of claim 1, wherein the cap is connected to the exhaust stack such that the cap is supported by the exhaust stack.

3. The exhaust handling system of claim 1, further comprising a vent, connected to the cap, wherein the vent also at least partially defines the first flow path, and wherein the vent includes at least one valve member positioned to operate between:

an open position to emit exhaust to the atmosphere via the first flow path, and

a closed position to prevent emission of exhaust to the atmosphere via the first flow path.

4. The exhaust handling system of claim 3, wherein the vent comprises a pressure-actuated vent.

5. The exhaust handling system of claim 4, wherein the at least one valve member comprises a plurality of louvers rotatable to transition the vent between the open position and the closed position.

6. The exhaust handling system of claim 1, wherein the coupling is positioned proximate to a deck of the marine vessel.

7. The exhaust handling system of claim 6, further comprising a cleanout port positioned along the collection pipe

25

proximate to the deck of the marine vessel, and wherein the cleanout port provides access into the collection pipe independent of the coupling.

8. The exhaust handling system of claim 1, further comprising a pipe connection positioned within the enclosure and in fluid communication with the exhaust pipe such that exhaust pipe is outputs exhaust into the pipe connection, and wherein the pipe connection is connected to the collection pipe such that exhaust emitted from the exhaust pipe is routed into the collection pipe via the pipe connection.

9. An exhaust handling system for a marine vessel, the exhaust handling system comprising:

a cap connected to a top end portion of an exhaust stack of the marine vessel such that the cap is supported by the exhaust stack and such that the cap defines an enclosure positioned to at least partially surround an outlet of an exhaust pipe extending through the exhaust stack of the marine vessel;

a pressure-actuated vent in fluid communication with the enclosure; and

a collection pipe in fluid communication with the enclosure to receive exhaust therefrom and connected to an exhaust cleaning assembly, the exhaust cleaning assembly including at least one tank to receive the exhaust.

10. The exhaust handling system of claim 9, wherein the pressure-actuated vent includes at least one valve member actuatable to selectively emit exhaust from the enclosure to an atmosphere surrounding the cap during operation.

11. The exhaust handling system of claim 9, wherein the pressure-actuated vent is positioned on a top end of the cap, and wherein the pressure-actuated vent is positioned along the collection pipe.

12. The exhaust handling system of claim 9, further comprising a coupling connected to the collection pipe and positioned to connect to a conduit of the exhaust cleaning assembly, wherein the coupling is accessible from a deck of the marine vessel, wherein the collection pipe comprises a cleanout port that is accessible from the deck of the marine vessel, and wherein the cleanout port is positioned to provide access into the collection pipe independent of the coupling.

13. The exhaust handling system of claim 9, further comprising a pipe connection positioned within the enclosure and in fluid communication with the exhaust pipe such that the exhaust pipe outputs exhaust to the collection pipe via the pipe connection during operation.

14. A method comprising:

(a) positioning a marine vessel in a berthing location, the marine vessel including a deck, an exhaust stack having a top end portion positioned above the deck, an exhaust pipe extending through the exhaust stack to an outlet and positioned above the top end portion of the exhaust stack, a cap connected to the exhaust stack so as to define an enclosure positioned to at least partially surround the outlet of the exhaust pipe, and a collection pipe in fluid communication with the enclosure to receive exhaust therefrom;

(b) connecting one or more tanks to the collection pipe after step (a), thereby to receive the exhaust; and

(c) operating the marine vessel to flow an exhaust emitted from the outlet of the exhaust pipe to the one or more tanks after step (b) and via the collection pipe.

26

15. The method of claim 14, further comprising:

(d) emitting the exhaust out of the enclosure through a vent to atmosphere surrounding the cap during step (a) when the marine vessel is in operation.

16. The method of claim 15, wherein step (d) comprises flowing the exhaust out of a pressure-actuated vent in fluid communication with the enclosure.

17. The method of claim 14, further comprising:

(e) disconnecting the exhaust cleaning assembly from the collection pipe;

(f) moving the marine vessel away from the berthing location after step (e); and

(g) flowing the exhaust out of the enclosure to atmosphere during step (f).

18. The method of claim 14, wherein step (b) comprises connecting a conduit connected to the one or more tanks to a coupling, the coupling connected to the collection pipe and positioned proximate to the deck of the marine vessel.

19. The method of claim 14, further comprising preventing the exhaust from flowing from the enclosure to atmosphere surrounding the cap during step (c).

20. A method comprising:

(a) positioning a marine vessel in a berthing location, the marine vessel including an exhaust stack, an exhaust pipe extending through the exhaust stack to an outlet positioned at an elevation above a top end portion of the exhaust stack, and a cap connected to the exhaust stack to define an enclosure positioned to surround the outlet of the exhaust pipe;

(b) flowing an exhaust from the enclosure to atmosphere surrounding the cap via a first flow path during step (a), the first flow path at least partially defined by the cap;

(c) connecting one or more tanks to receive the exhaust; and

(d) flowing the exhaust from the enclosure to the one or more tanks via a second flow path after step (b), the second flow path at least partially defined by the cap.

21. The method of claim 20, wherein step (b) further comprises flowing the exhaust out of a vent connected to the enclosure, and wherein the vent includes at least one louver, and wherein step (b) further comprises rotating the at least one louver of the vent to open the vent.

22. The method of claim 20, further comprising:

(e) disconnecting the exhaust cleaning assembly from the cap;

(f) moving the marine vessel away from the berthing location after step (e); and

(g) flowing the exhaust out of the enclosure to the atmosphere via the first flow path during step (f).

23. The method of claim 20, wherein step (c) comprises connecting a conduit connected to the one or more tanks to a coupling positioned proximate to a deck of the marine vessel and in fluid communication with the enclosure, and the method further comprising preventing the exhaust from flowing from the enclosure to the atmosphere during step (d).

24. The method of claim 20, wherein the one or more tanks is positioned on a barge positioned adjacent the berthing location.

* * * * *